

SIMATS ENGINEERING**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES****CHENNAI – 602105****Department of Computer Science and Engineering****ASSESSMENT TOOL 3 – OPEN DATA ANALYSIS PROJECT**

Course Code:	DSA05	Course Title:	Query Processing for Data Science in Real Time Applications
CO Assessed:	CO4	Assessment:	Tool 3 – Open Data Analysis Project
Weightage:		Total Marks:	50 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.. (BL4, BL5 and BL6)			
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Section / Batch:	2024	Date of Submission:	12 – 08 – 2026
Faculty In charge:	Dr. R.Mahesh Muthulakshmi	Assessment Mode:	Individual Submission

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _____

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

Road Accident Analysis and Traffic Safety Evaluation

1. Introduction

Road accidents are a major public safety concern worldwide, leading to loss of life, injuries, and economic damage. Understanding the factors contributing to road accidents can help authorities develop effective traffic safety measures. This project analyzes a road accident dataset to identify patterns related to accident severity, weather conditions, driver experience, vehicle involvement, and accident causes. The objective is to generate meaningful insights and recommend strategies for improving road safety.

2. Data Collection and Understanding

Dataset Source

The dataset was obtained from Kaggle and contains road accident records with information about drivers, vehicles, environmental conditions, and accident severity.

Dataset Description

The dataset consists of 12,316 accident records and 32 attributes. It includes information such as accident severity, day of the week, weather conditions, vehicle type, driver experience, number of casualties, and causes of accidents.

```

[1] import pandas as pd
df = pd.read_csv("../content/Road.csv")
print(df.shape)
df.head()

```

(5993, 32)

	Time	Day_of_week	Age_band_of_driver	Sex_of_driver	Educational_level	Vehicle_driver_relation	Driving_experience	Type_of_vehicle	Owner_of_vehicle	Service_year_of_vehicle	...	Vehicle_movement	Casualty_class	Sex_of_casualty
0	17:02:00	Monday	18-30	Male	Above high school	Employee	1-2yr	Automobile	Owner	Above 10yr	...	Going straight	na	
1	17:02:00	Monday	31-50	Male	Junior high school	Employee	Above 10yr	Public (> 45 seats)	Owner	5-10yrs	...	Going straight	na	
2	17:02:00	Monday	18-30	Male	Junior high school	Employee	1-2yr	Lorry (417100Q)	Owner	NaN	...	Going straight	Driver or rider	
3	1:06:00	Sunday	18-30	Male	Junior high school	Employee	5-10yr	Public (> 45 seats)	Governmental	NaN	...	Going straight	Pedestrian	F
4	1:06:00	Sunday	18-30	Male	Junior high school	Employee	2-5yr	NaN	Owner	5-10yrs	...	Going straight	na	

5 rows × 32 columns

Screenshot 1: Dataset Preview

```

[3] df.info()

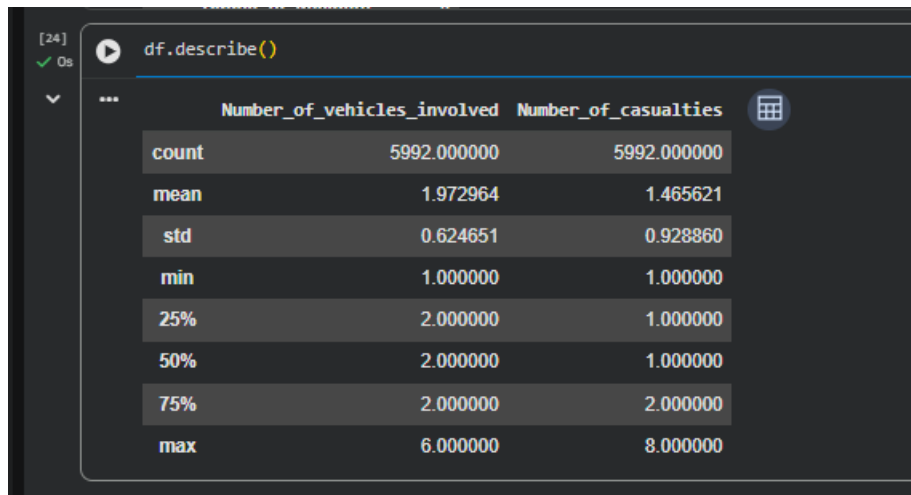
```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5993 entries, 0 to 5992
Data columns (total 32 columns):

#	Column	Non-Null Count	Dtype
0	Time	5993 non-null	object
1	Day_of_week	5993 non-null	object
2	Age_band_of_driver	5993 non-null	object
3	Sex_of_driver	5992 non-null	object
4	Educational_level	5643 non-null	object
5	Vehicle_driver_relation	5746 non-null	object
6	Driving_experience	5595 non-null	object
7	Type_of_vehicle	5511 non-null	object
8	Owner_of_vehicle	5762 non-null	object
9	Service_year_of_vehicle	4019 non-null	object
10	Defect_of_vehicle	3791 non-null	object
11	Area_accident_occured	5874 non-null	object
12	Lanes_or_Medians	5793 non-null	object
13	Road_alignment	5925 non-null	object
14	Types_of_Junction	5992 non-null	object
15	Road_surface_type	5911 non-null	object
16	Road_surface_conditions	5992 non-null	object
17	Light_conditions	5992 non-null	object
18	Weather_conditions	5992 non-null	object
19	Type_of_collision	5927 non-null	object
20	Number_of_vehicles_involved	5992 non-null	float64
21	Number_of_casualties	5992 non-null	float64
22	Vehicle_movement	5870 non-null	object
23	Casualty_class	5992 non-null	object
24	Sex_of_casualty	5992 non-null	object
25	Age_band_of_casualty	5992 non-null	object
26	Casualty_severity	5992 non-null	object
27	Work_of_casualty	4430 non-null	object
28	Fitness_of_casualty	4692 non-null	object
29	Pedestrian_movement	5992 non-null	object
30	Cause_of_accident	5992 non-null	object
31	Accident_severity	5992 non-null	object

dtypes: float64(2), object(30)
memory usage: 1.5+ MB

Screenshot 2: Dataset Information



	Number_of_vehicles_involved	Number_of_casualties
count	5992.000000	5992.000000
mean	1.972964	1.465621
std	0.624651	0.928860
min	1.000000	1.000000
25%	2.000000	1.000000
50%	2.000000	1.000000
75%	2.000000	2.000000
max	6.000000	8.000000

Screenshot 3: Dataset Statistics

Objectives

- Analyze accident severity patterns.
- Identify factors contributing to road accidents.
- Study the impact of weather and environmental conditions.
- Examine the relationship between driver experience and accident occurrence.
- Provide recommendations to improve traffic safety.

3. Data Cleaning and Preparation

The dataset was cleaned and prepared before analysis.

[4] `df.isnull().sum()`

...	0
Time	0
Day_of_week	0
Age_band_of_driver	0
Sex_of_driver	1
Educational_level	350
Vehicle_driver_relation	247
Driving_experience	398
Type_of_vehicle	482
Owner_of_vehicle	231
Service_year_of_vehicle	1974
Defect_of_vehicle	2202
Area_accident_occured	119
Lanes_or_Medians	200
Road_alignment	68
Types_of_Junction	1
Road_surface_type	82
Road_surface_conditions	1
Light_conditions	1
Weather_conditions	1
Type_of_collision	66
Number_of_vehicles_involved	1
Number_of_casualties	1
Vehicle_movement	123
Casualty_class	1
Sex_of_casualty	1
Age_band_of_casualty	1
Casualty_severity	1
Work_of_casualty	1563

Screenshot 4: Missing Values Analysis

[25] `print(df.shape)`
`df = df.drop_duplicates()`
`print(df.shape)`

... (5993, 32)
 (5993, 32)

Screenshot 5: Duplicate Removal

[26] ✓ Os df.dtypes

	0
Time	object
Day_of_week	object
Age_band_of_driver	object
Sex_of_driver	object
Educational_level	object
Vehicle_driver_relation	object
Driving_experience	object
Type_of_vehicle	object
Owner_of_vehicle	object
Service_year_of_vehicle	object
Defect_of_vehicle	object
Area_accident_occured	object
Lanes_or_Medians	object
Road_alignment	object
Types_of_Junction	object
Road_surface_type	object
Road_surface_conditions	object
Light_conditions	object
Weather_conditions	object
Type_of_collision	object
Number_of_vehicles_involved	float64
Number_of_casualties	float64

Screenshot 6: Data Type Conversion

The following preprocessing steps were performed:

- Imported the dataset into Google Colab using Pandas.
- Checked dataset structure using info() and head() functions.
- Identified missing values using isnull().sum().
- Removed duplicate records using drop_duplicates().

- Replaced missing categorical values with "Unknown".
- Converted numerical columns into appropriate numeric data types.
- Verified the dataset for consistency before analysis.

These preprocessing steps ensured that the dataset was suitable for exploratory data analysis and visualization.

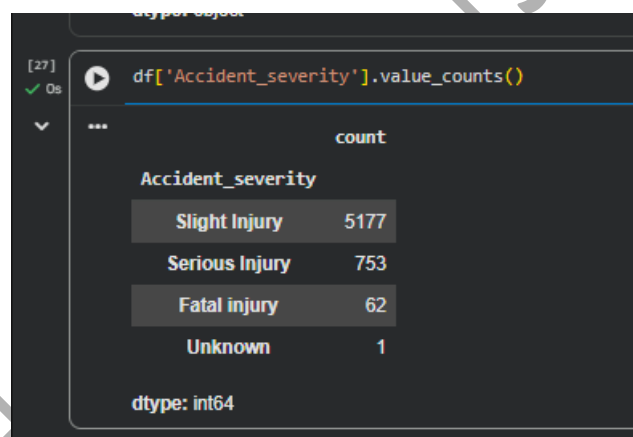
4. Exploratory Data Analysis

4.1 Accident Severity Analysis

Accident severity distribution was analyzed to understand the proportion of slight, serious, and fatal accidents.

Observation

The majority of accidents belong to the slight severity category, while serious and fatal accidents occur less frequently.



```
[27] df['Accident_severity'].value_counts()
```

Accident_severity	count
Slight Injury	5177
Serious Injury	753
Fatal injury	62
Unknown	1

dtype: int64



Interpretation

This indicates that although most accidents are not fatal, they still contribute significantly to injuries and property damage.

4.2 Day-wise Accident Analysis

The number of accidents occurring on different days of the week was analyzed.

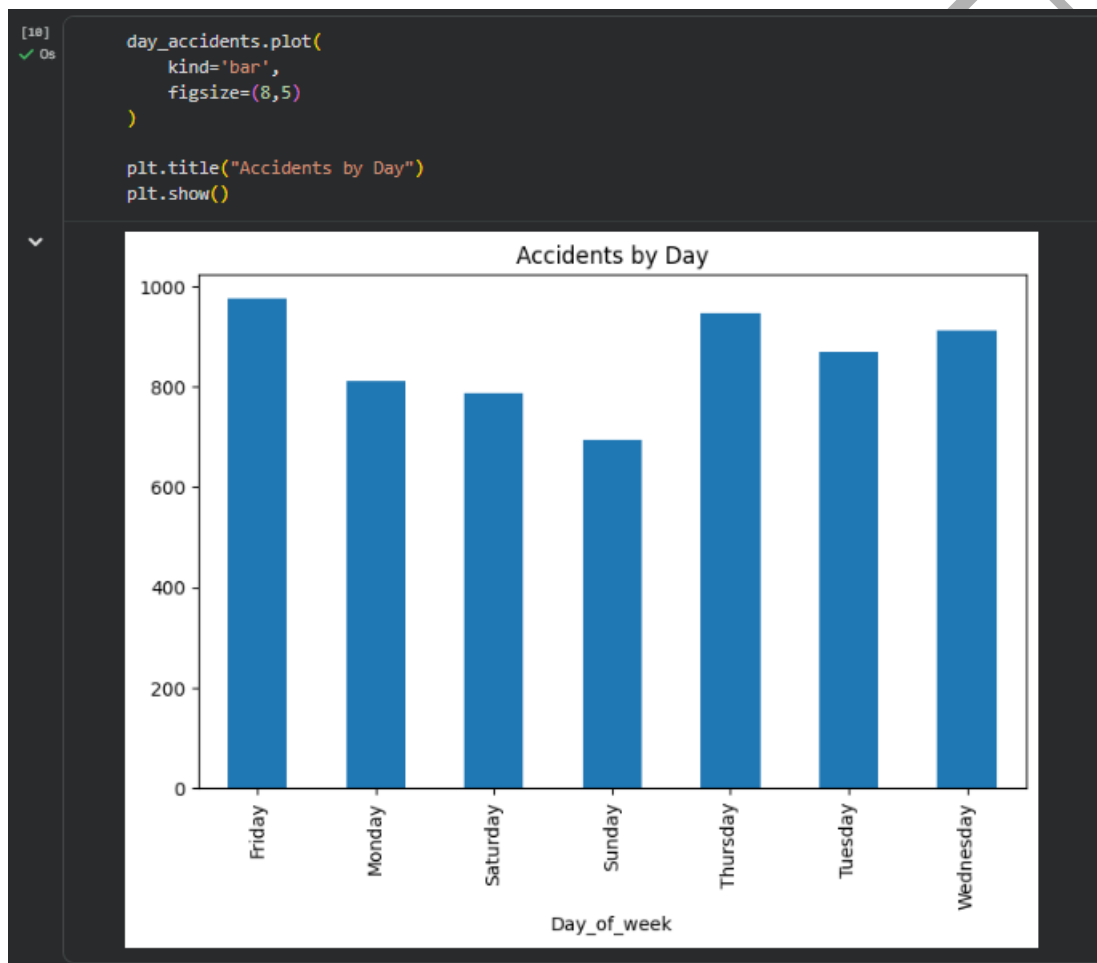
Observation

Certain days recorded a higher number of accidents compared to others.


```
[9]
✓ Os      day_accidents = df.groupby(
          'Day_of_week'
          ).size()

          print(day_accidents)

          Day_of_week
          Friday      975
          Monday      810
          Saturday     787
          Sunday       693
          Thursday     947
          Tuesday      869
          Wednesday    912
          dtype: int64
```



Interpretation

Traffic volume and commuting patterns may influence accident frequency on specific days.

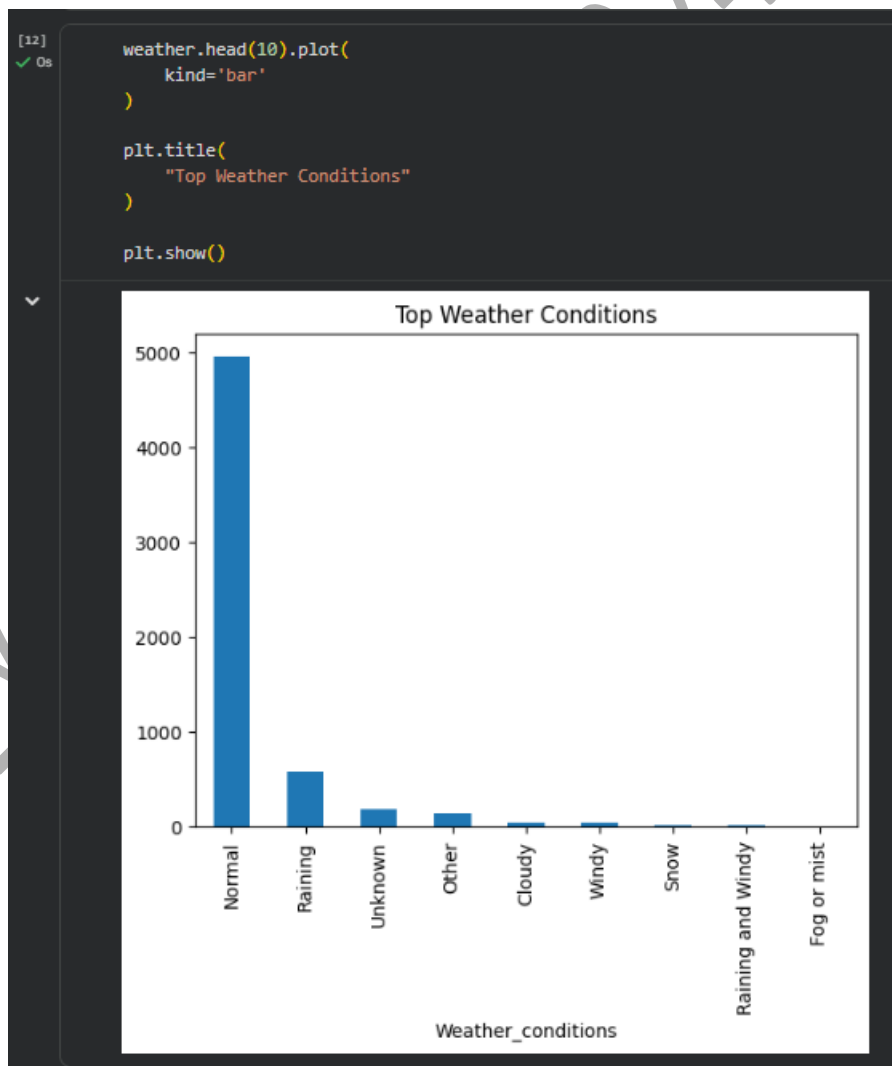
4.3 Weather Condition Analysis

Weather conditions during accidents were examined to identify environmental influences.

Observation

Normal weather conditions accounted for a large number of accidents, while adverse weather conditions such as rain and fog also contributed significantly.

```
[11] weather = df.groupby('Weather_conditions')\
      .size().sort_values(ascending=False)\
      .print()\
      print(weather)\
      Weather_conditions\n      Normal            4948\n      Raining           578\n      Unknown           191\n      Other             139\n      Cloudy            48\n      Windy             43\n      Snow              21\n      Raining and Windy  20\n      Fog or mist        5\n      dtype: int64
```



Interpretation

Although poor weather increases driving risk, a large number of accidents occur during normal weather due to higher traffic volumes.

4.4 Driver Experience Analysis

Driver experience was analyzed to study its relationship with accident occurrence.

Observation

Accidents were recorded across all experience levels.

```
[13] ✓ Os  experience = df.groupby(  
    'Driving_experience'  
).size().sort_values(  
    ascending=False  
)  
  
print(experience)
```

Driving_experience	
5-10yr	1665
2-5yr	1254
Above 10yr	1111
1-2yr	841
Below 1yr	646
Unknown	398
No Licence	60
unknown	18

dtype: int64

Interpretation

While inexperienced drivers may face higher risks, experienced drivers are also involved in accidents due to exposure and driving frequency.

4.5 Vehicle Type Analysis

Different vehicle categories involved in accidents were examined.

Observation

Certain vehicle types appeared more frequently in accident records than others.

```
[14] vehicle = df.groupby('Type_of_vehicle')
      .size().sort_values(ascending=False)
      print(vehicle.head(10))

*** Type_of_vehicle
Automobile      1573
Lorry (41?100Q) 1051
Other           566
Unknown         482
Pick up upto 10Q 383
Public (12 seats) 354
Stationwagen    333
Lorry (11?40Q)  265
Public (13?45 seats) 259
Public (> 45 seats) 204
dtype: int64
```

Interpretation

Vehicle usage patterns and traffic density contribute to differences in accident involvement.

4.6 Accident Cause Analysis

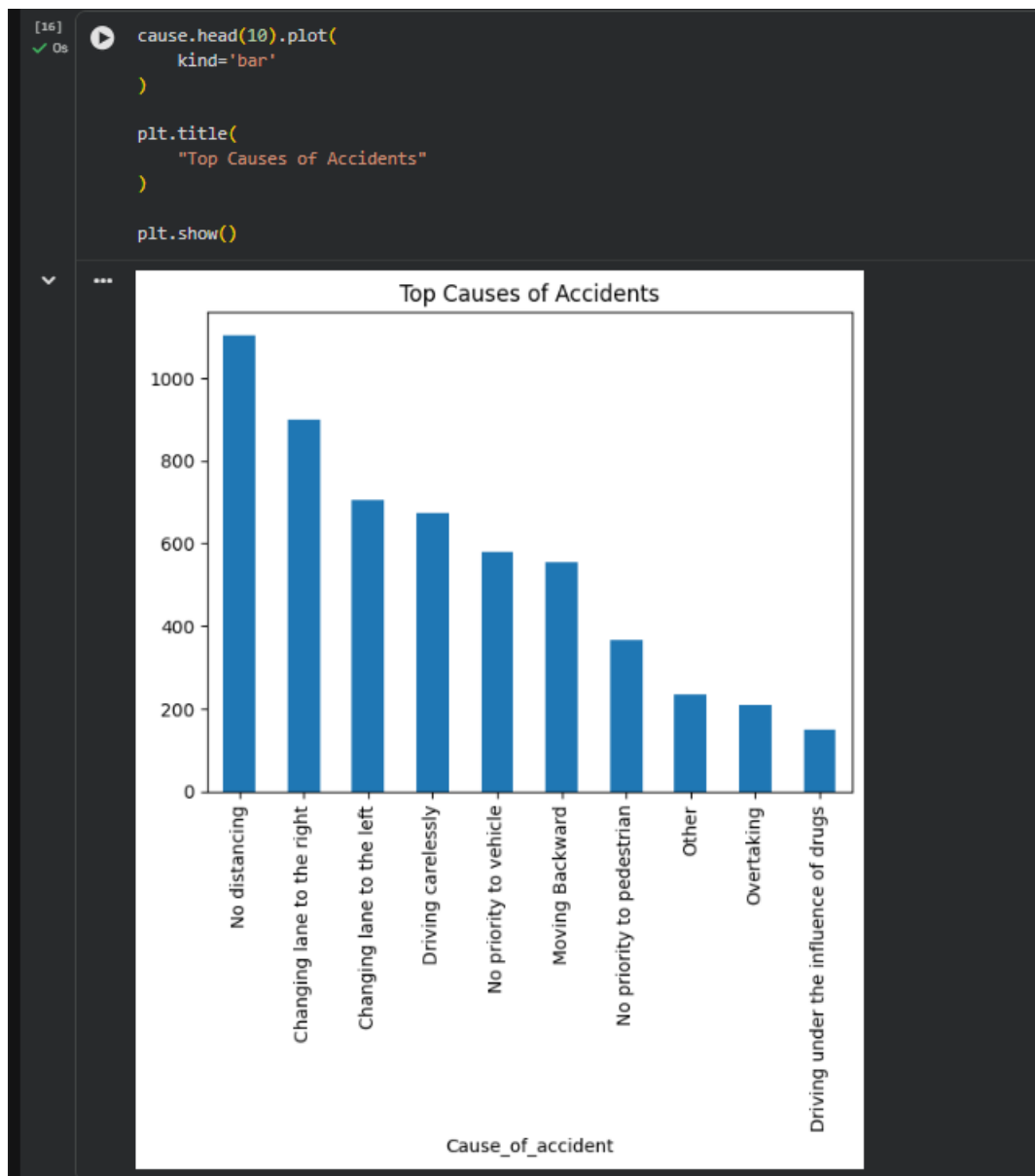
The causes of accidents were investigated.

Observation

The leading causes included driver-related factors such as speeding, improper driving behavior, and failure to maintain safe driving practices.

```
[15] cause = df.groupby('Cause_of_accident')
      .size().sort_values(ascending=False)
      print(cause.head(10))

*** Cause_of_accident
No distancing      1104
Changing lane to the right 898
Changing lane to the left 704
Driving carelessly  672
No priority to vehicle 579
Moving Backward     553
No priority to pedestrian 365
Other               235
Overtaking          208
Driving under the influence of drugs 151
dtype: int64
```



Interpretation

Human factors remain one of the most significant contributors to road accidents.

4.7 Light Condition Analysis

Light conditions at the time of accidents were analyzed.

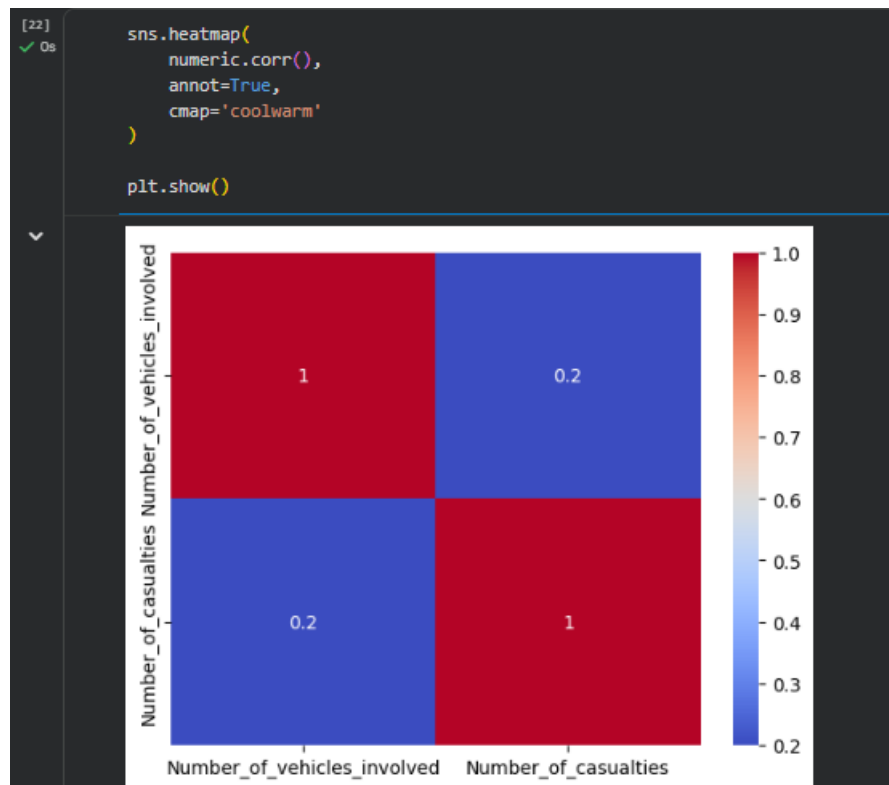
Observation

A large proportion of accidents occurred during daylight conditions, while a significant number also occurred at night.

```
[17] light = df.groupby('Light_conditions').size()

print(light)

... Light_conditions
Darkness - lights lit      1693
Darkness - lights unlit    11
Darkness - no lighting     83
Daylight                  4205
Unknown                    1
dtype: int64
```



Interpretation

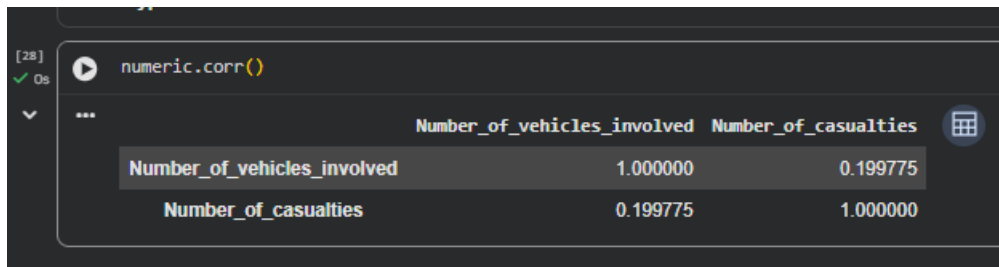
Night-time visibility and reduced driver alertness may increase accident risks.

5. Correlation Analysis

Correlation analysis was performed using numerical variables such as:

- Number of Vehicles Involved
- Number of Casualties

A correlation matrix and heatmap were generated to examine relationships between variables.



The screenshot shows a Jupyter Notebook cell with the code `numeric.corr()` executed. Below the code, a correlation matrix is displayed for two variables: `Number_of_vehicles_involved` and `Number_of_casualties`. The matrix shows a positive correlation of approximately 0.199775 between the two variables.

	Number_of_vehicles_involved	Number_of_casualties
Number_of_vehicles_involved	1.000000	0.199775
Number_of_casualties	0.199775	1.000000

Observation

The analysis revealed a positive relationship between the number of vehicles involved and the number of casualties.

Interpretation

Accidents involving multiple vehicles tend to result in a greater number of casualties.

6. Visualization

Several visualizations were created to support the analysis:

- Accident Severity Distribution Chart
- Accidents by Day of the Week
- Weather Condition Analysis
- Driver Experience Distribution
- Vehicle Type Analysis
- Top Causes of Accidents
- Correlation Heatmap

These visualizations helped identify trends and communicate findings effectively.

7. Key Findings

1. Most accidents were classified as slight severity accidents.
2. Accident frequency varied across days of the week.
3. Weather conditions influenced accident occurrence and severity.
4. Driver behavior was a major contributing factor.
5. Multiple-vehicle accidents were associated with higher casualty counts.
6. Both daytime and nighttime accidents were significant.

7. Vehicle type and driving experience showed notable patterns in accident involvement.
-

8. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Increase road safety awareness programs focusing on responsible driving behavior.
 2. Strengthen enforcement against speeding and traffic violations.
 3. Improve road infrastructure and warning systems in accident-prone areas.
 4. Enhance visibility measures such as improved street lighting and road markings.
 5. Promote defensive driving training for drivers of all experience levels.
 6. Implement weather-based traffic alerts during adverse conditions.
 7. Increase monitoring and surveillance in high-risk traffic zones.
-

9. Conclusion

This project successfully analyzed road accident data to identify important factors influencing accident occurrence and severity. The findings indicate that driver behavior, environmental conditions, vehicle involvement, and road conditions significantly affect traffic safety. Through data cleaning, exploratory analysis, visualization, and correlation analysis, meaningful insights were obtained that can support informed decision-making and contribute to reducing road accidents and improving public safety.

Air Quality and Environmental Impact Assessment

1. Introduction

Air pollution has become one of the most significant environmental challenges affecting human health and sustainable development. Rapid urbanization, industrialization, increasing vehicular emissions, and population growth have contributed to the deterioration of air quality in many cities. Poor air quality can lead to respiratory diseases, cardiovascular disorders, and other serious health issues.

This project aims to analyze air quality data collected from various Indian cities to understand pollution patterns, identify highly polluted regions, examine the relationship between different pollutants, and evaluate their impact on the Air Quality Index (AQI). Various analytical and visualization techniques are used to generate meaningful insights and support environmental management decisions.

2. Data Collection and Understanding

Dataset Source

The dataset was obtained from Kaggle and contains air quality measurements collected from multiple Indian cities.

Dataset Description

The dataset contains daily air quality observations from various cities and includes important pollutants such as:

- PM2.5
- PM10
- NO
- NO2
- NOx
- NH3
- CO
- SO2
- O3
- Benzene
- Toluene

- Xylene
- AQI
- AQI_Bucket

The dataset consists of approximately 29,531 records and 16 attributes.

Project Objectives

- Analyze air pollution levels across cities.
- Study AQI distribution and pollution categories.
- Identify highly polluted cities.
- Examine relationships between pollutants and AQI.
- Explore monthly pollution trends.
- Provide recommendations for environmental improvement.

3. Data Cleaning and Preparation

Data preprocessing was performed before conducting analysis.

The following steps were carried out:

Data Inspection

The dataset structure was examined using the following functions:

- `df.head()`

```
[3]
✓ Os
import pandas as pd

df = pd.read_csv("/content/city_day.csv")

print(df.shape)
df.head()
```

(29531, 16)

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	NaN	NaN
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	NaN	NaN
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	NaN	NaN
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	NaN	NaN
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	NaN	NaN

- `df.info()`

```
[5] df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   City         29531 non-null  object
1   Date         29531 non-null  object
2   PM2.5        24933 non-null  float64
3   PM10         18391 non-null  float64
4   NO           25949 non-null  float64
5   NO2          25946 non-null  float64
6   NOx          25346 non-null  float64
7   NH3          19203 non-null  float64
8   CO           27472 non-null  float64
9   SO2          25677 non-null  float64
10  O3           25509 non-null  float64
11  Benzene      23908 non-null  float64
12  Toluene      21490 non-null  float64
13  Xylene       11422 non-null  float64
14  AQI          24850 non-null  float64
15  AQI_Bucket   24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB
```

- df.describe()

df.describe()

	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	Month
count	29531	24933.000000	18391.000000	25949.000000	25946.000000	25346.000000	19203.000000	27472.000000	25677.000000	25509.000000	23908.000000	21490.000000	11422.000000	24850.000000	29531.000000
mean	2019-05-14 05:40:15.807118000	67.450578	118.127103	17.574730	28.560659	32.309123	23.483476	2.248598	14.531977	34.491430	3.280840	8.700972	3.070128	166.463581	6.267143
min	2015-01-01 00:00:00	0.040000	0.010000	0.020000	0.010000	0.000000	0.010000	0.000000	0.010000	0.010000	0.000000	0.000000	0.000000	13.000000	1.000000
25%	2017-04-16 00:00:00	28.820000	56.255000	5.630000	11.750000	12.820000	8.580000	0.510000	5.670000	18.860000	0.120000	0.600000	0.140000	81.000000	3.000000
50%	2018-08-05 00:00:00	48.570000	95.680000	9.880000	21.690000	23.520000	15.850000	0.890000	9.160000	30.840000	1.070000	2.970000	0.980000	118.000000	6.000000
75%	2019-09-03 00:00:00	80.590000	149.745000	19.950000	37.620000	40.127500	30.020000	1.450000	15.220000	45.570000	3.080000	9.150000	3.350000	208.000000	9.000000
max	2020-07-01 00:00:00	949.990000	1000.000000	390.680000	362.210000	467.630000	352.890000	175.810000	193.860000	257.730000	455.030000	454.850000	170.370000	2049.000000	12.000000
std	NaN	64.661449	90.605110	22.785846	24.474746	31.646911	25.684275	6.962884	18.133775	21.694928	15.811136	19.969164	6.323247	140.696585	3.431974

Missing Value Analysis

Missing values were identified using:

df.isnull().sum()

Several pollutant attributes contained missing values that required attention.

```
[6] df.isnull().sum()
```

...	0
City	0
Date	0
PM2.5	4598
PM10	11140
NO	3582
NO2	3585
NOx	4185
NH3	10328
CO	2059
SO2	3854
O3	4022
Benzene	5623
Toluene	8041
Xylene	18109
AQI	4681
AQI_Bucket	4681

dtype: int64

Duplicate Removal

Duplicate records were identified and removed to improve data quality.

```
[7] df.duplicated().sum()
np.int64(0)

[8] df = df.drop_duplicates()
```

Data Type Conversion

The Date column was converted into datetime format for time-series analysis.

```
[9] df['Date'] = pd.to_datetime(df['Date'])
```

Feature Engineering

A new Month column was created from the Date attribute to analyze monthly pollution trends.

```
[10] df['Month'] = df['Date'].dt.month
```

These preprocessing steps improved the quality and reliability of the dataset for further analysis.

4. Exploratory Data Analysis

4.1 AQI Category Distribution Analysis

The AQI_Bucket attribute was analyzed to understand the distribution of air quality categories.

Observation

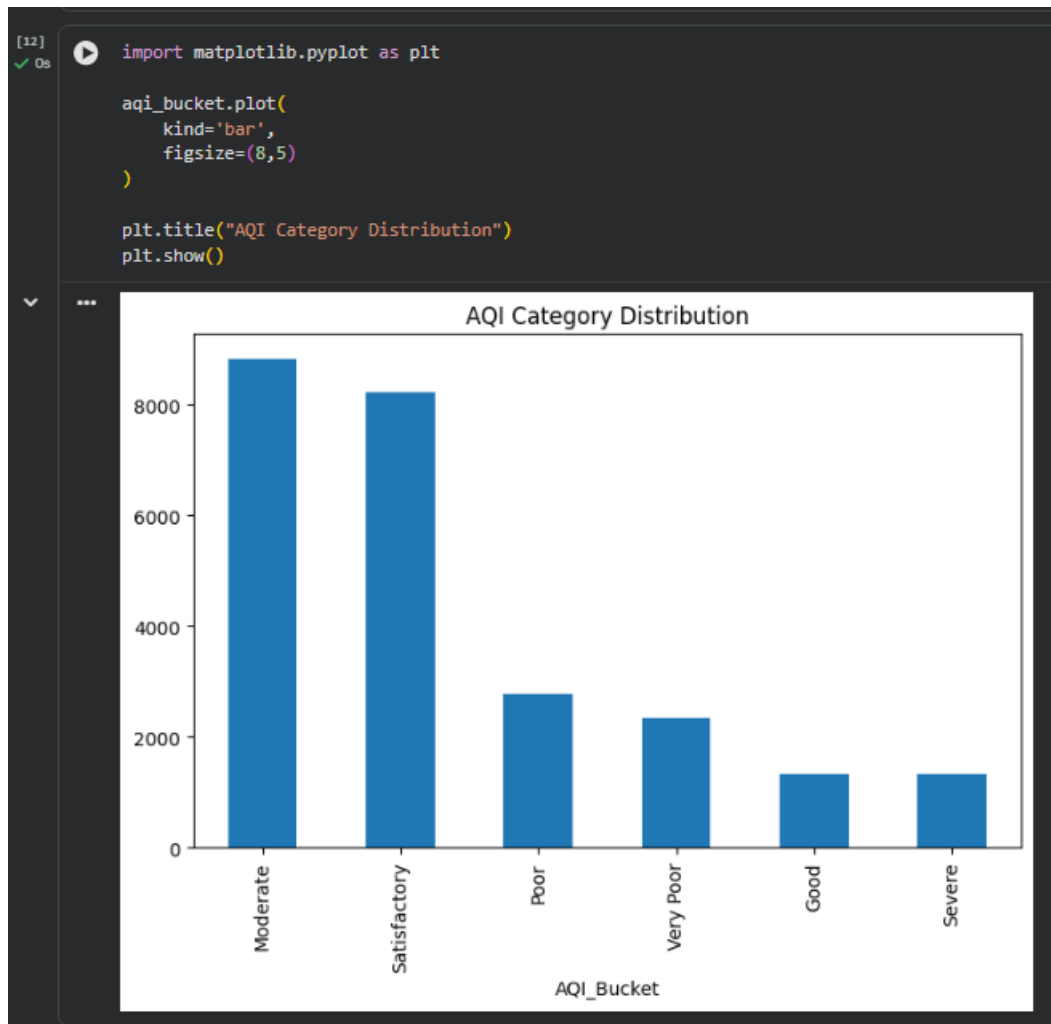
The dataset contains multiple AQI categories including:

- Good
- Satisfactory
- Moderate
- Poor
- Very Poor
- Severe

Interpretation

```
[11]
✓ Os      aqi_bucket = df['AQI_Bucket'].value_counts()
          print(aqi_bucket)

AQI_Bucket
Moderate      8829
Satisfactory  8224
Poor          2781
Very Poor    2337
Good          1341
Severe        1338
Name: count, dtype: int64
```



Most observations fall under Moderate and Poor categories, indicating that air pollution remains a significant concern in several cities.

4.2 Top 10 Most Polluted Cities Analysis

Average AQI values were calculated for each city and ranked accordingly.

Observation

Certain cities consistently recorded higher AQI values compared to others.

Interpretation

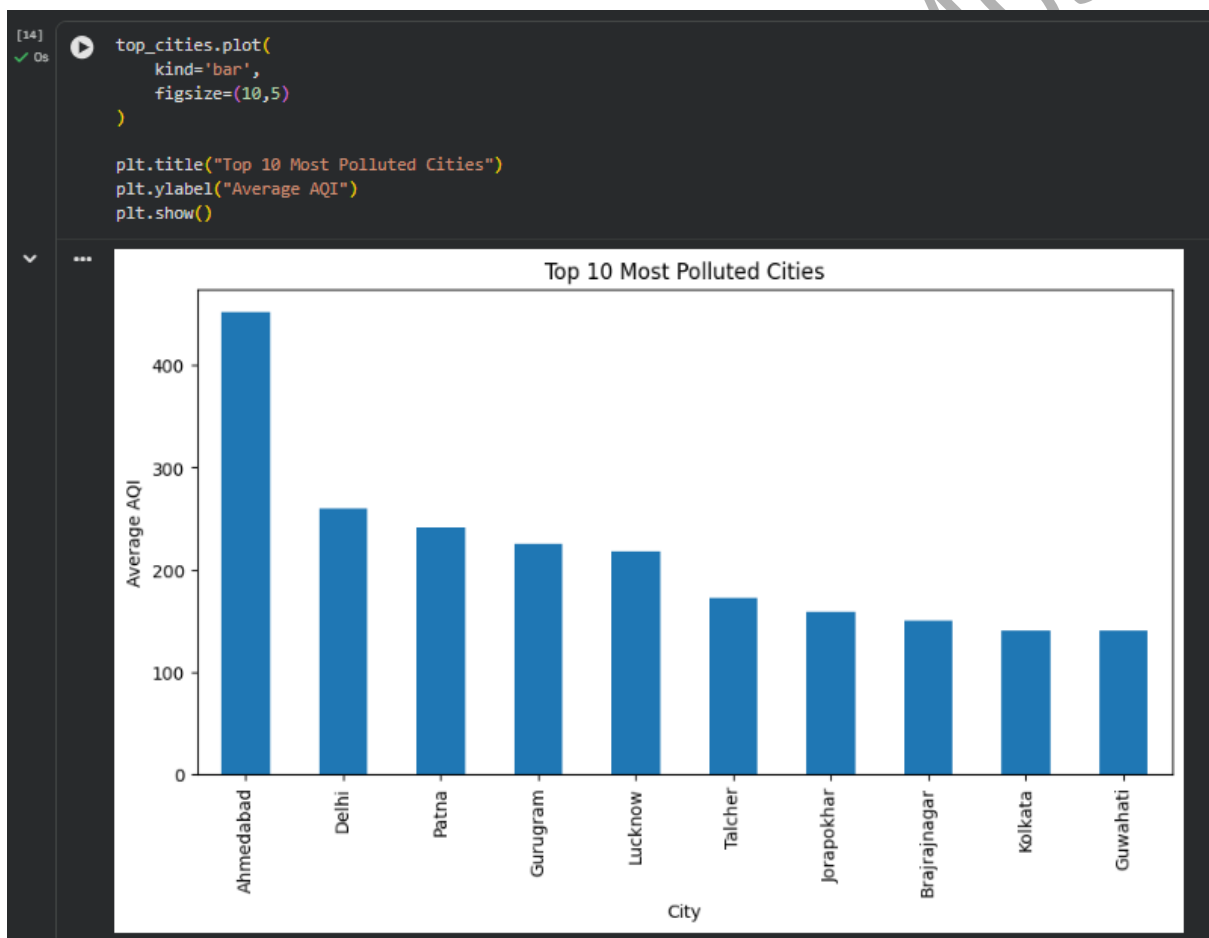
These cities may experience higher industrial activity, traffic congestion, and population density, contributing to poor air quality.

```
[13] top_cities = df.groupby('City')
      ['AQI'].mean().sort_values(
      ascending=False)
      .head(10)

      print(top_cities)
```

City	AQI
Ahmedabad	452.122939
Delhi	259.487744
Patna	240.782042
Gurugram	225.123882
Lucknow	217.973059
Talcher	172.886819
Jorapokhar	159.251621
Brajrajnagar	150.280505
Kolkata	140.566313
Guwahati	140.111111

Name: AQI, dtype: float64



4.3 Monthly AQI Trend Analysis

Monthly average AQI values were analyzed to identify seasonal pollution trends.

Observation

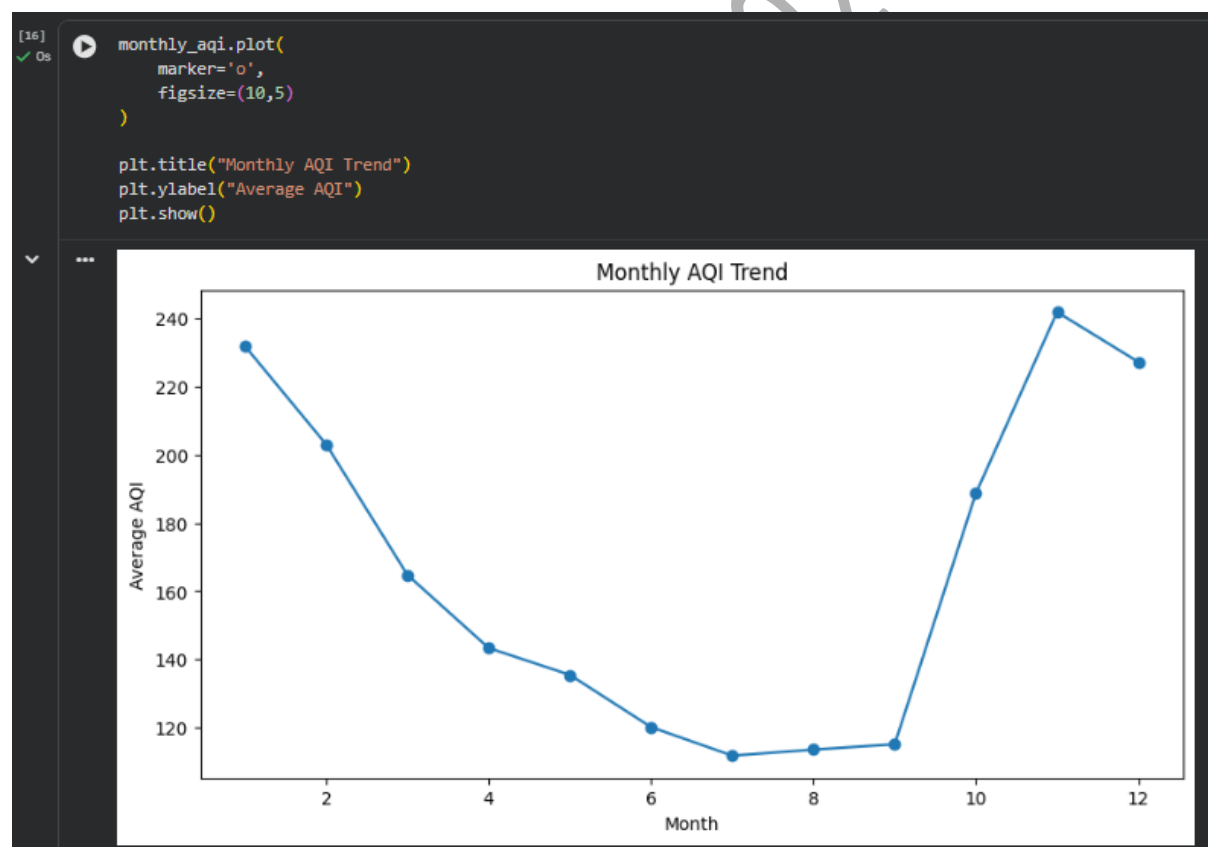
AQI levels varied throughout the year.

Interpretation

Seasonal weather conditions, industrial emissions, and vehicular activities influence pollution concentrations across different months.

```
[15] monthly_aqi = df.groupby('Month')
      ['AQI'].mean()
      print(monthly_aqi)

... Month
1      231.674918
2      202.905197
3      164.735281
4      143.355120
5      135.489579
6      120.198379
7      111.854575
8      113.613176
9      115.191804
10     188.613552
11     241.681302
12     227.084980
      Name: AQI, dtype: float64
```



4.4 PM2.5 and AQI Relationship Analysis

A scatter plot was created to study the relationship between PM2.5 concentration and AQI.

Observation

Higher PM2.5 concentrations generally corresponded to higher AQI values.

Interpretation



PM2.5 is one of the major pollutants affecting overall air quality and public health.

4.5 Pollutant Concentration Analysis

Average concentrations of major pollutants were compared.

Pollutants analyzed include:

- PM2.5
- PM10
- NO2

- SO2
- CO
- O3

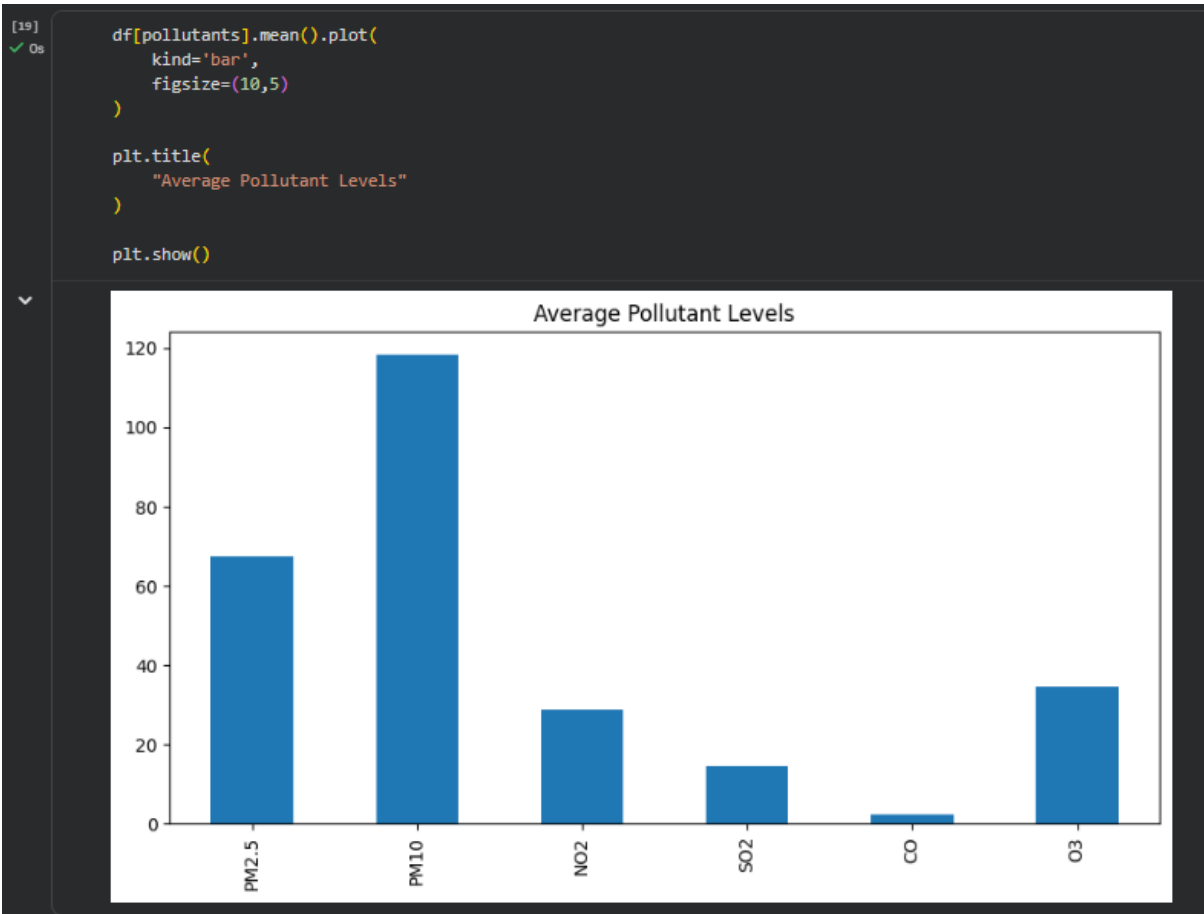
Observation

Particulate pollutants such as PM2.5 and PM10 showed higher concentrations than several gaseous pollutants.

```
[18] ✓ 0s pollutants = [  
    'PM2.5',  
    'PM10',  
    'NO2',  
    'SO2',  
    'CO',  
    'O3'  
]  
  
df[pollutants].mean()  
  
... 0  
PM2.5 67.450578  
PM10 118.127103  
NO2 28.560659  
SO2 14.531977  
CO 2.248598  
O3 34.491430  
  
dtype: float64
```

Interpretation

Particulate matter significantly contributes to environmental pollution and AQI deterioration.



4.6 Correlation Analysis

Correlation analysis was performed among major pollutants and AQI.

Observation

Strong positive correlations were observed between AQI and particulate pollutants.

[28]

0s

corr_columns = [

'PM2.5',

'PM10',

'NO',

'NO2',

'NOx',

'NH3',

'CO',

'SO2',

'O3',

'AQI'

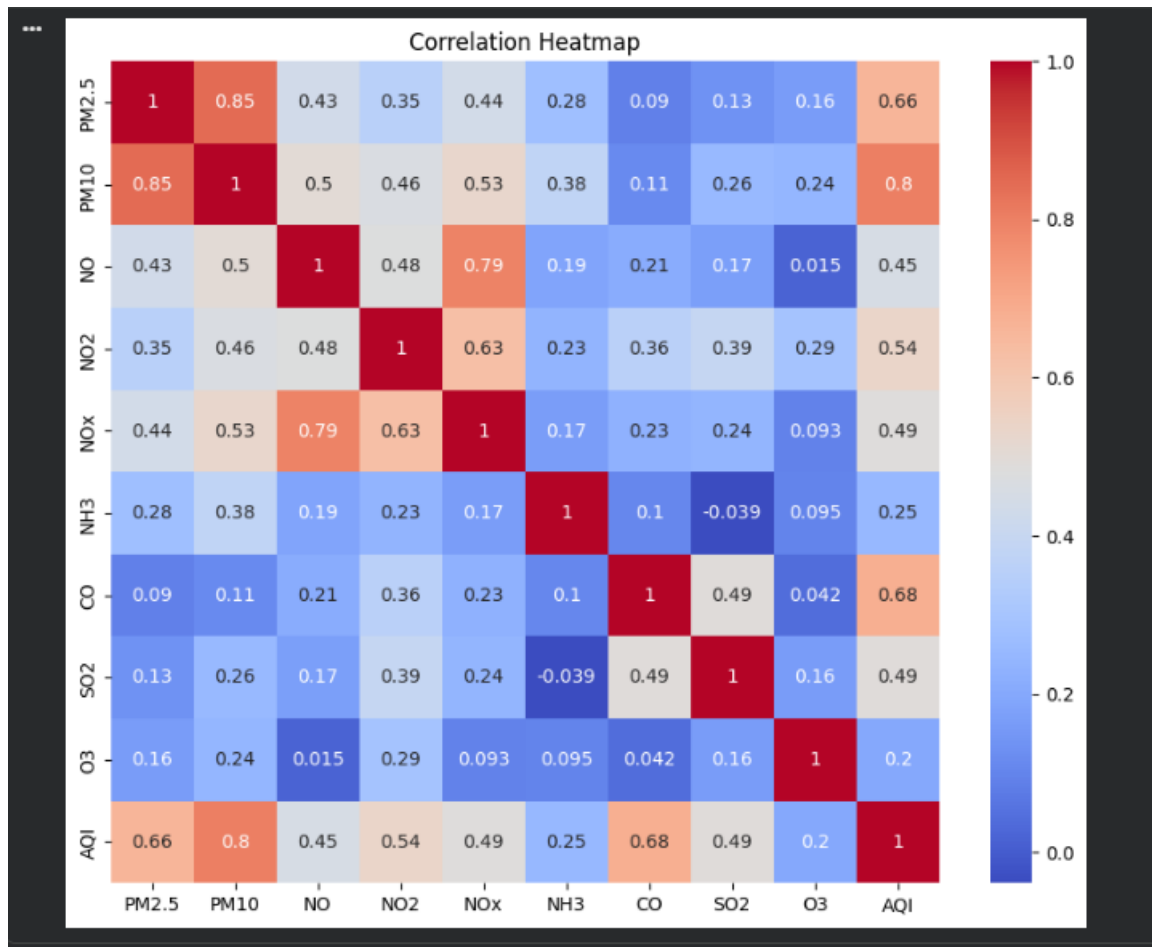
]

df[corr_columns].corr()

	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	AQI
PM2.5	1.000000	0.846498	0.433491	0.350709	0.436792	0.275086	0.089912	0.132325	0.161238	0.659181
PM10	0.846498	1.000000	0.502349	0.464380	0.527768	0.376816	0.112588	0.256974	0.244919	0.803313
NO	0.433491	0.502349	1.000000	0.478070	0.794890	0.185621	0.212607	0.170322	0.014580	0.452191
NO2	0.350709	0.464380	0.478070	1.000000	0.627627	0.234938	0.356521	0.392233	0.293349	0.537071
NOx	0.436792	0.527768	0.794890	0.627627	1.000000	0.166224	0.226992	0.238397	0.093170	0.486450
NH3	0.275086	0.376816	0.185621	0.234938	0.166224	1.000000	0.104891	-0.038998	0.094972	0.252019
CO	0.089912	0.112588	0.212607	0.356521	0.226992	0.104891	1.000000	0.489697	0.041736	0.683346
SO2	0.132325	0.256974	0.170322	0.392233	0.238397	-0.038998	0.489697	1.000000	0.162142	0.490586
O3	0.161238	0.244919	0.014580	0.293349	0.093170	0.094972	0.041736	0.162142	1.000000	0.198991
AQI	0.659181	0.803313	0.452191	0.537071	0.486450	0.252019	0.683346	0.490586	0.198991	1.000000

Interpretation

An increase in pollutant concentration generally leads to deterioration in air quality.



4.7 Top 10 Cities by PM2.5 Analysis

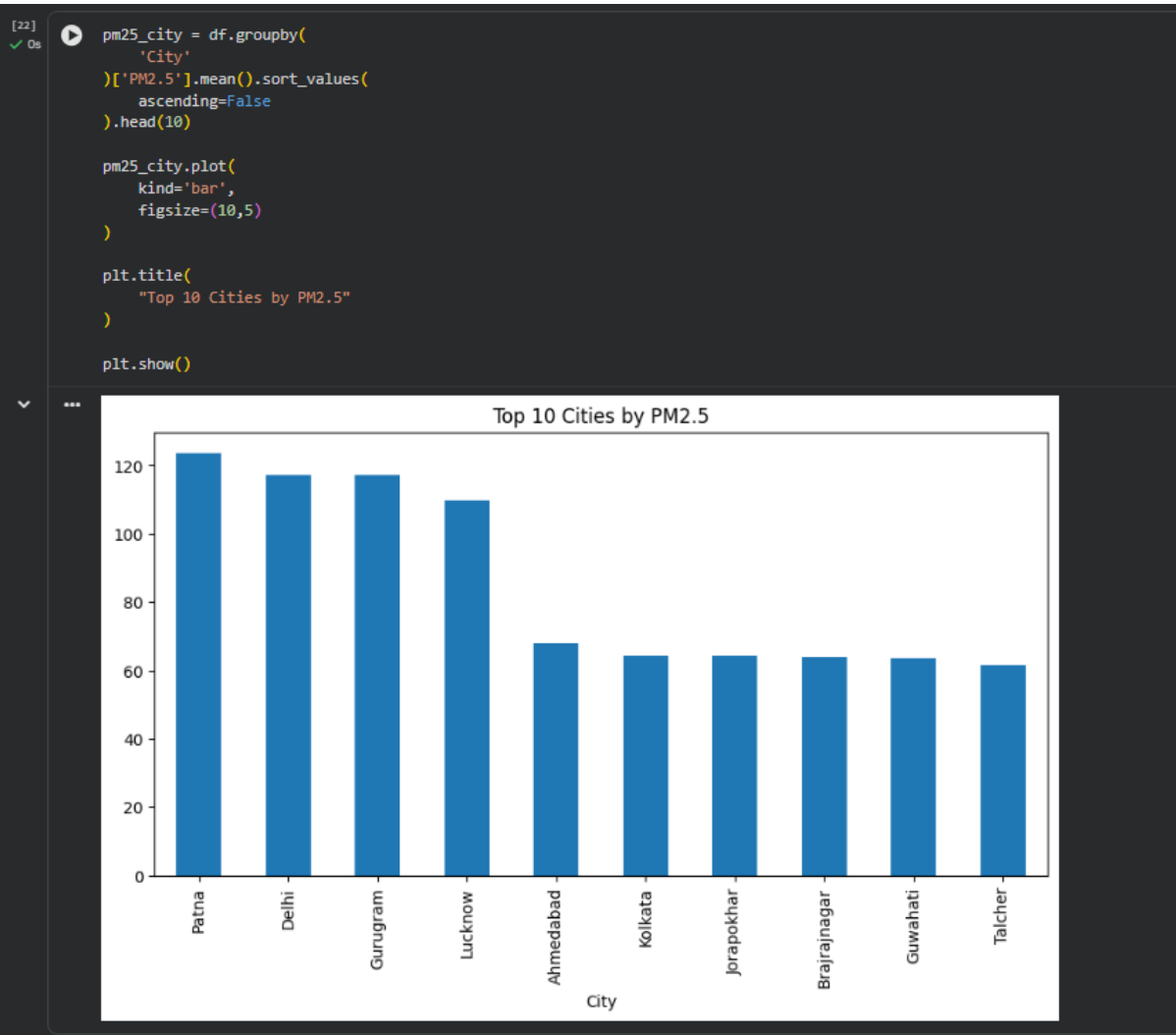
Cities were ranked based on average PM2.5 concentration.

Observation

Several cities recorded extremely high PM2.5 values.

Interpretation

These locations may require stricter pollution control measures and environmental monitoring.



5. Visualization and Dashboard Design

Several visualizations were developed to support data interpretation.

Visualizations Created

1. AQI Category Distribution
2. Top 10 Most Polluted Cities
3. Monthly AQI Trend
4. PM2.5 vs AQI Scatter Plot
5. Average Pollutant Comparison
6. Correlation Heatmap
7. Top 10 Cities by PM2.5

Dashboard Components

KPI Cards

- Total Records
- Average AQI
- Highest AQI
- Number of Cities

Charts

- AQI Distribution
- Polluted Cities Analysis
- Monthly AQI Trend
- Pollutant Comparison
- Correlation Heatmap

The dashboard provides a comprehensive overview of pollution trends and environmental conditions.

6. Key Findings

1. Air pollution levels varied significantly across cities.
 2. Several cities consistently recorded Poor and Very Poor AQI categories.
 3. PM2.5 and PM10 were the major contributors to air pollution.
 4. AQI values showed noticeable seasonal variations.
 5. Strong positive correlations existed between AQI and particulate pollutants.
 6. Certain cities experienced considerably higher pollution levels than others.
 7. Pollutant concentration directly influenced overall air quality conditions.
-

7. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Strengthen air quality monitoring systems in highly polluted cities.
2. Promote the use of cleaner fuels and renewable energy sources.

3. Increase urban afforestation and green infrastructure projects.
 4. Encourage public transportation and reduce private vehicle dependency.
 5. Implement stricter industrial emission standards.
 6. Conduct regular environmental awareness programs.
 7. Develop city-specific pollution control strategies.
 8. Enhance environmental monitoring during high-pollution seasons.
-

8. Conclusion

This project successfully analyzed air quality data from multiple cities to evaluate pollution levels and environmental impact. Through data cleaning, exploratory data analysis, correlation analysis, and visualization techniques, significant patterns and trends were identified. The results indicate that particulate matter pollutants, particularly PM2.5 and PM10, play a major role in determining air quality. The insights obtained from this study can assist policymakers, environmental agencies, and public health organizations in developing effective strategies to improve air quality and promote sustainable urban development.

GLOBAL EDUCATION AND LITERACY PERFORMANCE STUDY

1. Introduction

Education is a key factor influencing economic growth, social development, and quality of life. Literacy rates, school enrollment, and educational attainment are important indicators used to evaluate a country's educational performance. This project analyzes global education data from multiple countries to examine literacy rates, enrollment levels, and their relationship with socioeconomic factors such as birth rate and unemployment rate. The analysis helps identify educational trends and provides insights for policy improvement.

2. Data Collection and Understanding

Dataset Source

Kaggle – Global Education Dataset

Dataset Description

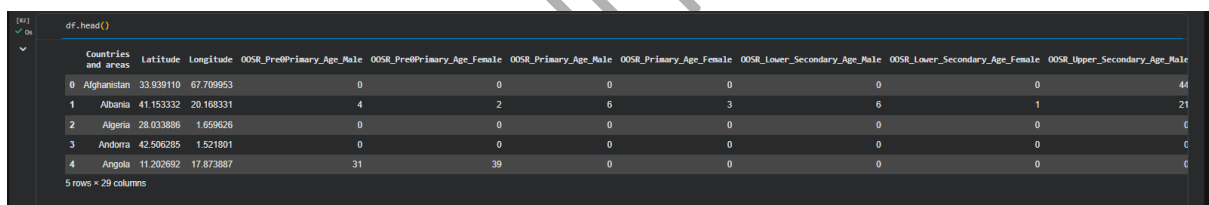
The dataset contains educational indicators from various countries around the world. It includes information related to:

- Male Literacy Rate
- Female Literacy Rate
- Gross Primary Education Enrollment
- Gross Tertiary Education Enrollment
- Birth Rate
- Unemployment Rate
- Country Information

The dataset consists of educational and socioeconomic indicators that help evaluate the educational status of different nations.

```
[81]
✓ Os      print(df.shape)
      (202, 29)
```

 **Screenshot 1: Dataset Shape Output (df.shape)**



	Countries and areas	Latitude	Longitude	00SR_Prc@Primary_Age_Male	00SR_Prc@Primary_Age_Female	00SR_Primary_Age_Male	00SR_Primary_Age_Female	00SR_Lower_Secondary_Age_Male	00SR_Lower_Secondary_Age_Female	00SR_Upper_Secondary_Age_Male
0	Afghanistan	33.939110	67.709953	0	0	0	0	0	0	44
1	Albania	41.153332	20.168331	4	2	6	3	6	1	21
2	Algeria	28.033886	1.659626	0	0	0	0	0	0	0
3	Andorra	42.506285	1.521801	0	0	0	0	0	0	0
4	Angola	11.202692	17.873887	31	39	0	0	0	0	0

5 rows x 29 columns

 **Screenshot 2: Dataset Preview (df.head())**

```
[83] df.info()
✓ Os

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 202 entries, 0 to 201
Data columns (total 29 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Countries and areas                                                    202 non-null   object
1   Latitude                                                              202 non-null   float64
2   Longitude                                                             202 non-null   float64
3   OOSR_Pre0Primary_Age_Male                                             202 non-null   int64
4   OOSR_Pre0Primary_Age_Female                                           202 non-null   int64
5   OOSR_Primary_Age_Male                                                 202 non-null   int64
6   OOSR_Primary_Age_Female                                               202 non-null   int64
7   OOSR_Lower_Secondary_Age_Male                                         202 non-null   int64
8   OOSR_Lower_Secondary_Age_Female                                       202 non-null   int64
9   OOSR_Upper_Secondary_Age_Male                                         202 non-null   int64
10  OOSR_Upper_Secondary_Age_Female                                       202 non-null   int64
11  Completion_Rate_Primary_Male                                           202 non-null   int64
12  Completion_Rate_Primary_Female                                         202 non-null   int64
13  Completion_Rate_Lower_Secondary_Male                                   202 non-null   int64
14  Completion_Rate_Lower_Secondary_Female                               202 non-null   int64
15  Completion_Rate_Upper_Secondary_Male                                   202 non-null   int64
16  Completion_Rate_Upper_Secondary_Female                               202 non-null   int64
17  Grade_2_3_Proficiency_Reading                                         202 non-null   int64
18  Grade_2_3_Proficiency_Math                                             202 non-null   int64
19  Primary_End_Proficiency_Reading                                         202 non-null   int64
20  Primary_End_Proficiency_Math                                           202 non-null   int64
21  Lower_Secondary_End_Proficiency_Reading                               202 non-null   int64
22  Lower_Secondary_End_Proficiency_Math                                   202 non-null   int64
23  Youth_15_24_Literacy_Rate_Male                                         202 non-null   int64
24  Youth_15_24_Literacy_Rate_Female                                       202 non-null   int64
25  Birth_Rate                                                             202 non-null   float64
26  Gross_Primary_Education_Enrollment                                    202 non-null   float64
27  Gross_Tertiary_Education_Enrollment                                    202 non-null   float64
28  Unemployment_Rate                                                      202 non-null   float64
dtypes: float64(6), int64(22), object(1)
memory usage: 45.9+ KB
```

 **Screenshot 3: Dataset Information (df.info())**

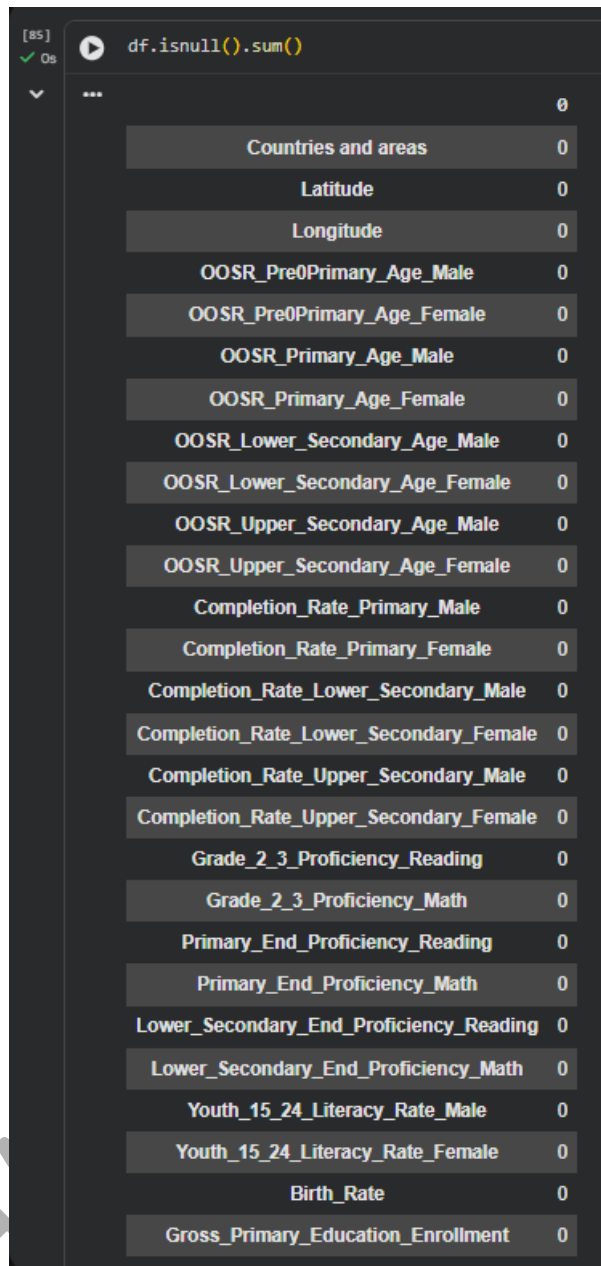
3. Data Cleaning and Preparation

Before performing analysis, the dataset was cleaned and prepared.

The following preprocessing steps were performed:

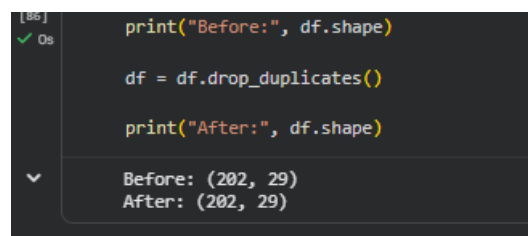
- Imported dataset into Google Colab.
- Examined dataset structure and attributes.
- Identified missing values.
- Converted educational indicators into numeric format.
- Removed invalid symbols and text values.
- Replaced missing values using mean imputation.
- Verified data quality before analysis.

These preprocessing steps improved data consistency and ensured reliable analytical results.



Variable	Sum
...	0
Countries and areas	0
Latitude	0
Longitude	0
OOSR_Pre0Primary_Age_Male	0
OOSR_Pre0Primary_Age_Female	0
OOSR_Primary_Age_Male	0
OOSR_Primary_Age_Female	0
OOSR_Lower_Secondary_Age_Male	0
OOSR_Lower_Secondary_Age_Female	0
OOSR_Upper_Secondary_Age_Male	0
OOSR_Upper_Secondary_Age_Female	0
Completion_Rate_Primary_Male	0
Completion_Rate_Primary_Female	0
Completion_Rate_Lower_Secondary_Male	0
Completion_Rate_Lower_Secondary_Female	0
Completion_Rate_Upper_Secondary_Male	0
Completion_Rate_Upper_Secondary_Female	0
Grade_2_3_Proficiency_Reading	0
Grade_2_3_Proficiency_Math	0
Primary_End_Proficiency_Reading	0
Primary_End_Proficiency_Math	0
Lower_Secondary_End_Proficiency_Reading	0
Lower_Secondary_End_Proficiency_Math	0
Youth_15_24_Literacy_Rate_Male	0
Youth_15_24_Literacy_Rate_Female	0
Birth_Rate	0
Gross_Primary_Education_Enrollment	0

 **Screenshot 4:** Missing Values Analysis (df.isnull().sum())



```
[85]
✓ Os
print("Before:", df.shape)

df = df.drop_duplicates()

print("After:", df.shape)

Before: (202, 29)
After: (202, 29)
```

 **Screenshot 5:** Data Cleaning Process

	Latitude	Longitude	OOSR_PrePrimary_Age_Male	OOSR_PrePrimary_Age_Female	OOSR_Primary_Age_Male	OOSR_Primary_Age_Female	OOSR_Lower_Secondary_Age_Male	OOSR_Lower_Secondary_Age_Female	OOSR_Upper_Secondary_Age_Male	OOSR_Upper_Secondary_Age_Female
count	202.000000	202.000000	202.000000	202.000000	202.000000	202.000000	202.000000	202.000000	202.000000	202.000000
mean	25.081422	55.166928	19.658416	19.282178	5.282178	5.569307	8.707921	8.831683	20.292079	20.292079
std	16.813639	45.976287	25.007604	25.171147	9.396442	10.383092	13.258203	14.724717	21.485592	21.485592
min	0.023559	0.824782	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	11.685062	18.665678	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.250000	0.250000
50%	21.207851	43.518091	9.000000	7.000000	1.000000	1.000000	2.000000	2.000000	15.000000	15.000000
75%	39.901792	77.684945	31.000000	30.000000	6.000000	6.750000	12.750000	10.750000	32.750000	32.750000
max	64.963051	178.065032	96.000000	96.000000	58.000000	67.000000	61.000000	70.000000	84.000000	84.000000

Screenshot 6: Statistical Summary (df.describe())

4. Exploratory Data Analysis

4.1 Male Literacy Rate Analysis

The literacy rates of males aged 15–24 years were analyzed across different countries.

Observation

Several countries recorded literacy rates close to 100%, indicating strong educational systems and widespread access to education.

```

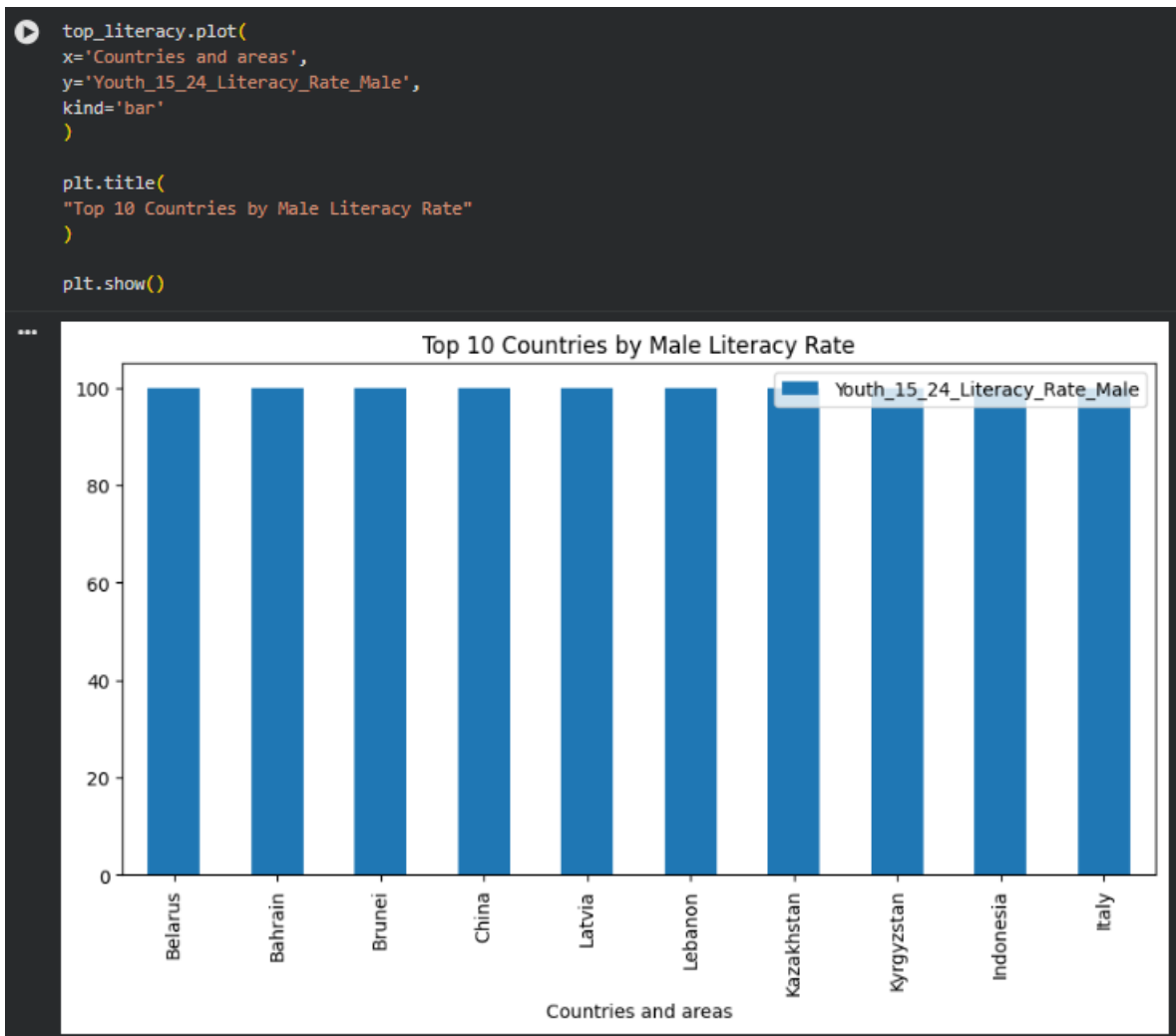
top_literacy = df[
    ['Countries and areas',
     'Youth_15_24_Literacy_Rate_Male']
].sort_values(
    'Youth_15_24_Literacy_Rate_Male',
    ascending=False
).head(10)

print(top_literacy)

```

	Countries and areas	Youth_15_24_Literacy_Rate_Male
16	Belarus	100
13	Bahrain	100
26	Brunei	100
37	China	100
97	Latvia	100
98	Lebanon	100
91	Kazakhstan	100
95	Kyrgyzstan	100
82	Indonesia	100
87	Italy	100

Screenshot 7: Top 10 Countries by Male Literacy Rate Table



 **Screenshot 8:** Male Literacy Rate Bar Chart

4.2 Female Literacy Rate Analysis

Female literacy rates were analyzed to understand educational accessibility among women.

Observation

Female literacy rates were generally high in developed countries and showed strong improvement across many regions.

```

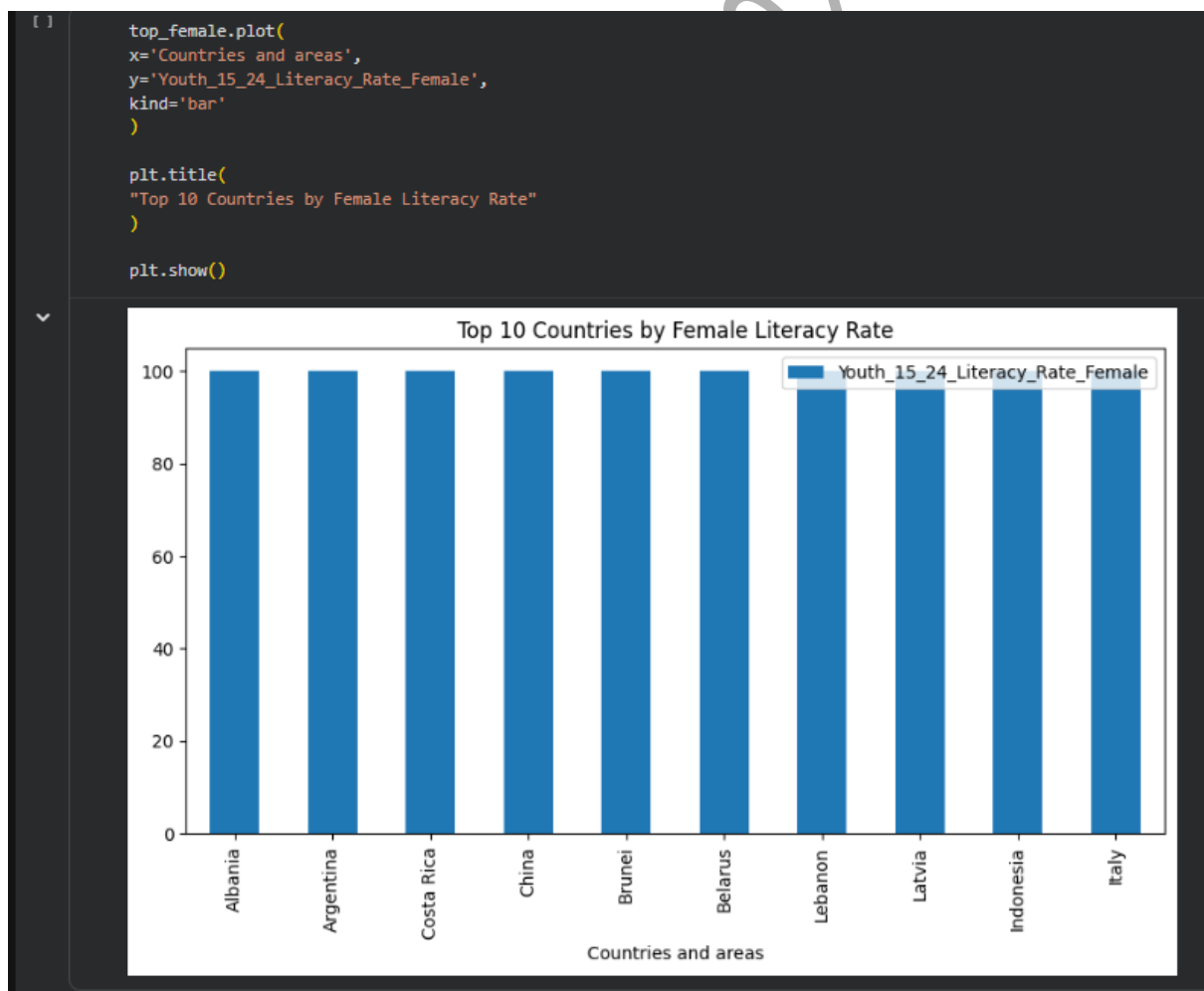
top_female = df[
    [
        'Countries and areas',
        'Youth_15_24_Literacy_Rate_Female'
    ]
].sort_values(
    'Youth_15_24_Literacy_Rate_Female',
    ascending=False
).head(10)

print(top_female)

```

	Countries and areas	Youth_15_24_Literacy_Rate_Female
1	Albania	100
7	Argentina	100
42	Costa Rica	100
37	China	100
26	Brunei	100
16	Belarus	100
98	Lebanon	100
97	Latvia	100
82	Indonesia	100
87	Italy	100

 **Screenshot 9: Top 10 Countries by Female Literacy Rate Table**



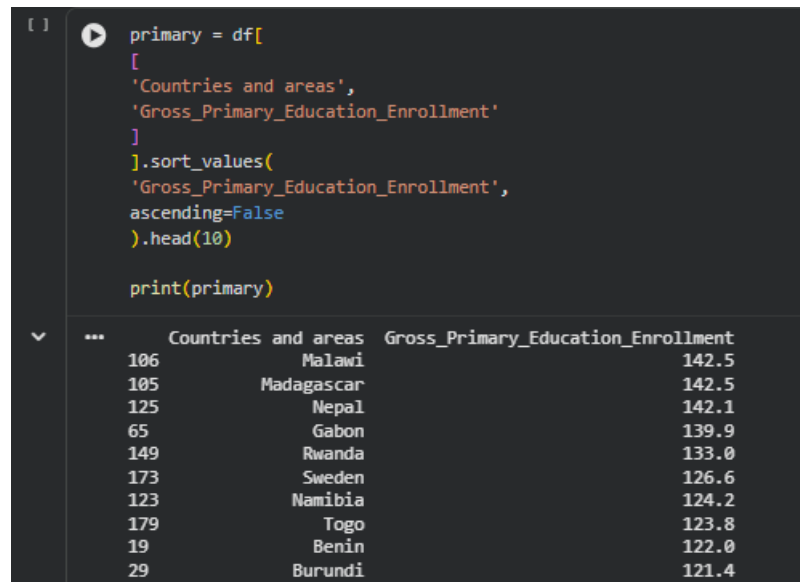
 **Screenshot 10: Female Literacy Rate Bar Chart**

4.3 Primary Education Enrollment Analysis

Primary school enrollment rates were examined to evaluate access to basic education.

Observation

Most countries exhibited high enrollment rates in primary education, demonstrating strong participation at the foundational education level.

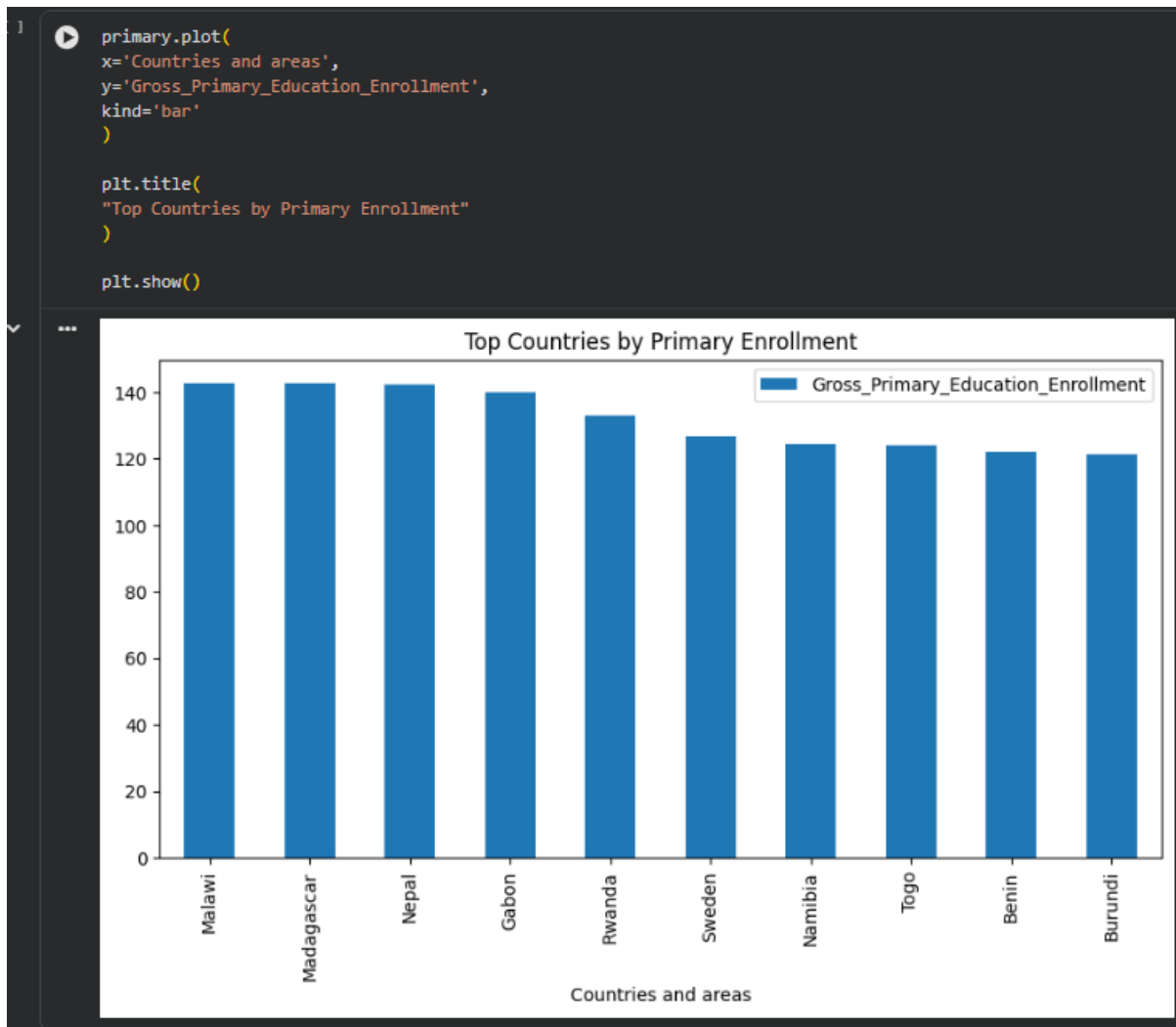


```
[ ] ▶ primary = df[
    [
        'Countries and areas',
        'Gross_Primary_Education_Enrollment'
    ]
].sort_values(
    'Gross_Primary_Education_Enrollment',
    ascending=False
).head(10)

print(primary)
```

	Countries and areas	Gross_Primary_Education_Enrollment
106	Malawi	142.5
105	Madagascar	142.5
125	Nepal	142.1
65	Gabon	139.9
149	Rwanda	133.0
173	Sweden	126.6
123	Namibia	124.2
179	Togo	123.8
19	Benin	122.0
29	Burundi	121.4

 **Screenshot 11:** Primary Enrollment Table



 **Screenshot 12:** Primary Enrollment Bar Chart

4.4 Tertiary Education Enrollment Analysis

Higher education enrollment rates were analyzed to understand participation in advanced education.

Observation

Significant variation was observed among countries, indicating differences in higher education accessibility and economic development.


```

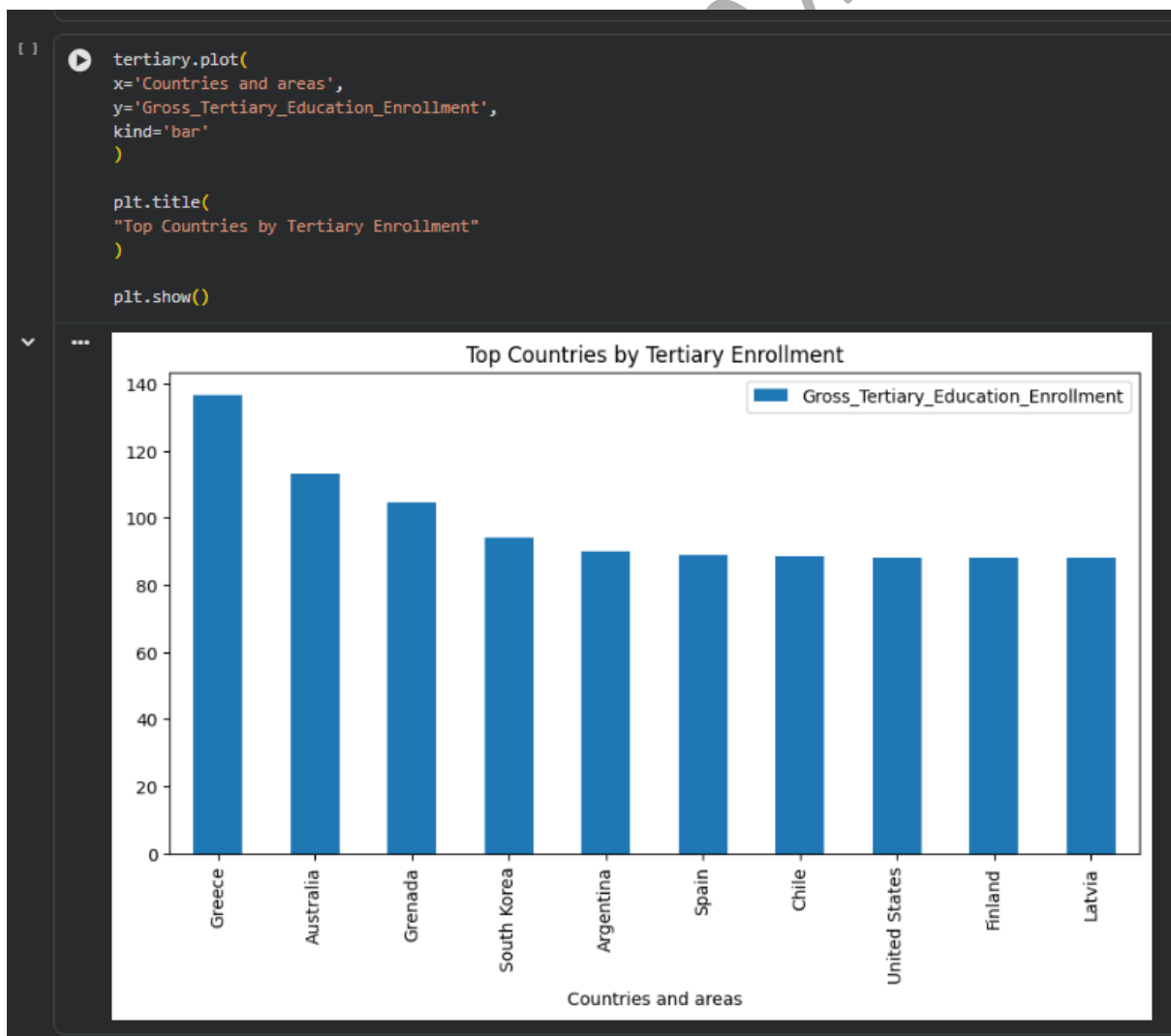
[ ] tertiary = df[
    [
        'Countries and areas',
        'Gross_Tertiary_Education_Enrollment'
    ]
].sort_values(
    'Gross_Tertiary_Education_Enrollment',
    ascending=False
).head(10)

print(tertiary)

```

	Countries and areas	Gross_Tertiary_Education_Enrollment
70	Greece	136.6
9	Australia	113.1
71	Grenada	104.6
145	South Korea	94.3
7	Argentina	90.0
168	Spain	88.9
36	Chile	88.5
193	United States	88.2
63	Finland	88.2
97	Latvia	88.1

 **Screenshot 13:** Tertiary Enrollment Table



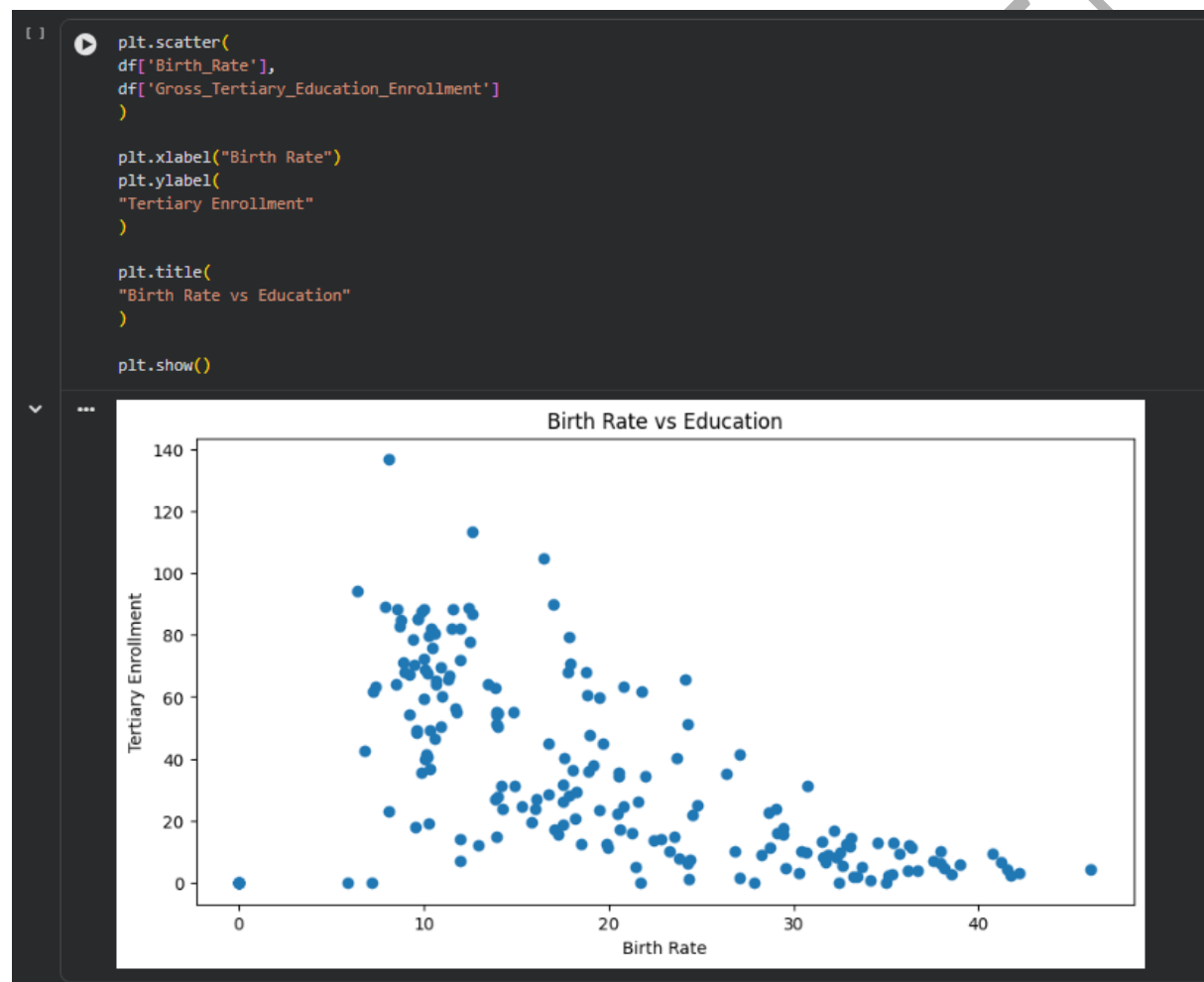
 **Screenshot 14:** Tertiary Enrollment Bar Chart

4.5 Birth Rate and Education Analysis

The relationship between birth rate and tertiary education enrollment was examined using a scatter plot.

Observation

Countries with higher educational participation generally tended to exhibit lower birth rates.



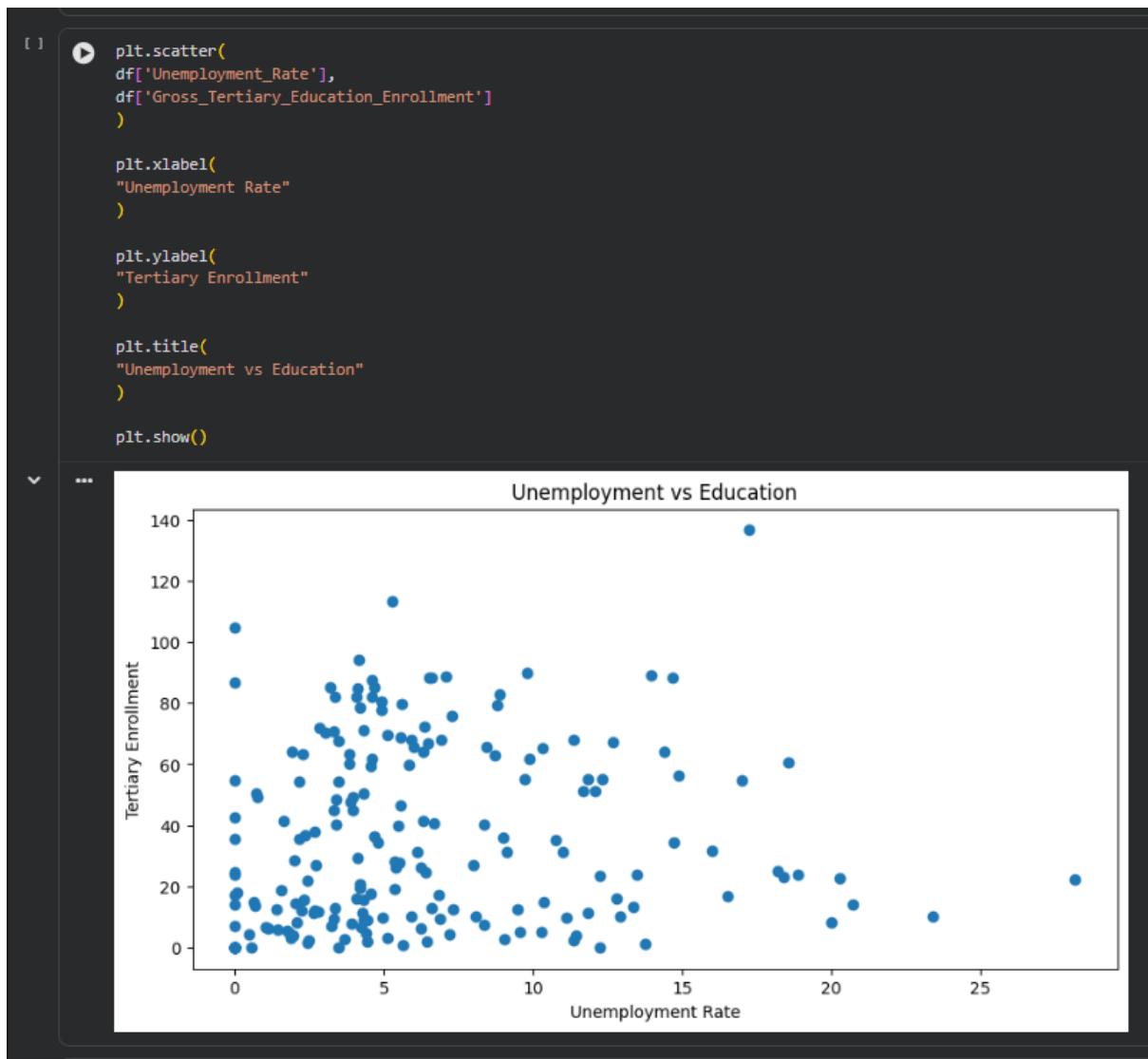
 **Screenshot 15:** Birth Rate vs Education Scatter Plot

4.6 Unemployment and Education Analysis

The relationship between unemployment rate and tertiary education enrollment was analyzed.

Observation

Educational attainment appeared to influence employment opportunities and workforce participation.



 **Screenshot 16:** Unemployment Rate vs Education Scatter Plot

4.7 Correlation Analysis

Correlation analysis was performed among key educational and socioeconomic indicators.

Variables included:

- Male Literacy Rate
- Female Literacy Rate
- Birth Rate

- Gross Primary Enrollment
- Gross Tertiary Enrollment
- Unemployment Rate

Observation

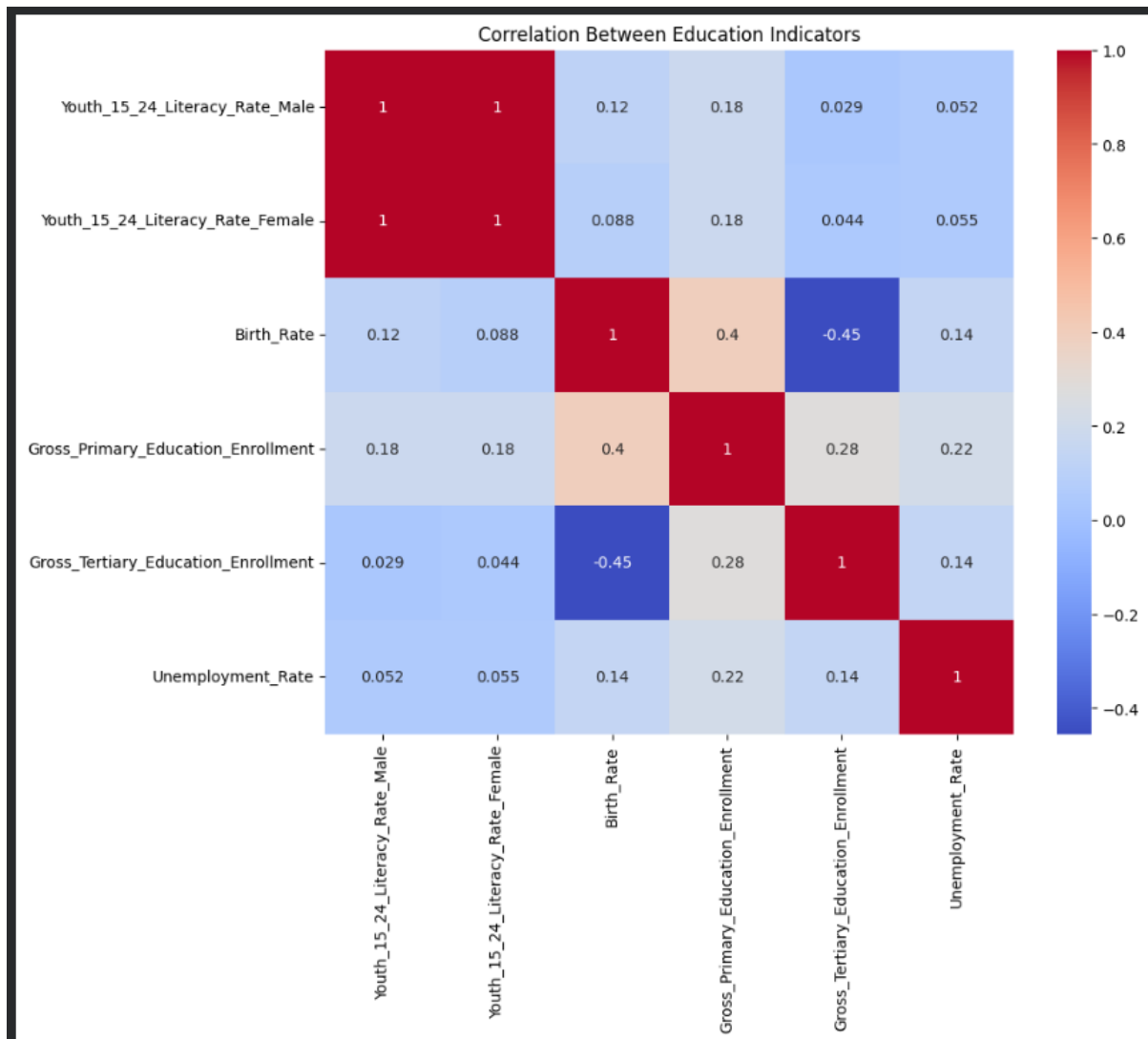
Strong relationships were identified among several educational indicators, suggesting that literacy and enrollment levels are closely linked.

```
corr_cols = [
    'Youth_15_24_Literacy_Rate_Male',
    'Youth_15_24_Literacy_Rate_Female',
    'Birth_Rate',
    'Gross_Primary_Education_Enrollment',
    'Gross_Tertiary_Education_Enrollment',
    'Unemployment_Rate'
]

df[corr_cols].corr()
```

	Youth_15_24_Literacy_Rate_Male	Youth_15_24_Literacy_Rate_Female	Birth_Rate	Gross_Primary_Education_Enrollment	Gross_Tertiary_Education_Enrollment	Unemployment_Rate
Youth_15_24_Literacy_Rate_Male	1.000000	0.996766	0.116816	0.176601	0.028501	0.052013
Youth_15_24_Literacy_Rate_Female	0.996766	1.000000	0.087788	0.176927	0.043842	0.055326
Birth_Rate	0.116816	0.087788	1.000000	0.395862	-0.454775	0.143130
Gross_Primary_Education_Enrollment	0.176601	0.176927	0.395862	1.000000	0.283959	0.215464
Gross_Tertiary_Education_Enrollment	0.028501	0.043842	-0.454775	0.283959	1.000000	0.140702
Unemployment_Rate	0.052013	0.055326	0.143130	0.215464	0.140702	1.000000

 **Screenshot 17: Correlation Matrix**



 **Screenshot 18:** Correlation Heatmap

5. Visualization and Dashboard Design

Various visualizations were developed to effectively communicate educational trends and patterns.

Visualizations Created

- Male Literacy Rate Bar Chart
- Female Literacy Rate Bar Chart
- Primary Enrollment Bar Chart
- Tertiary Enrollment Bar Chart
- Birth Rate vs Education Scatter Plot

- Unemployment vs Education Scatter Plot
- Correlation Heatmap

The visualizations provide a clear understanding of educational performance and its association with socioeconomic factors.

6. Key Findings

1. Several countries achieved literacy rates above 95%.
 2. Female literacy rates were comparable to male literacy rates in many countries.
 3. Primary education enrollment was significantly higher than tertiary enrollment.
 4. Access to higher education varied considerably across nations.
 5. Educational indicators showed meaningful relationships with socioeconomic factors.
 6. Countries with strong educational systems generally demonstrated better development outcomes.
 7. Literacy and enrollment rates remain important indicators of national progress.
-

7. Recommendations

1. Increase investment in educational infrastructure.
 2. Promote equal educational opportunities for all genders.
 3. Improve access to higher education in developing nations.
 4. Strengthen literacy programs in low-performing regions.
 5. Encourage vocational and skill-based education.
 6. Develop policies that improve educational accessibility and affordability.
 7. Enhance educational awareness programs to reduce dropout rates.
-

8. Conclusion

This project successfully analyzed global education indicators using exploratory data analysis and visualization techniques. The study revealed important patterns in literacy rates, enrollment levels, and socioeconomic factors across different countries. The findings emphasize the importance of education in promoting economic growth,

reducing inequality, and improving quality of life. Effective educational policies and continuous investment in learning opportunities are essential for sustainable national development.

JEYA HARSHINI R 192424051